



Faculty of Computers and Artificial Intelligence

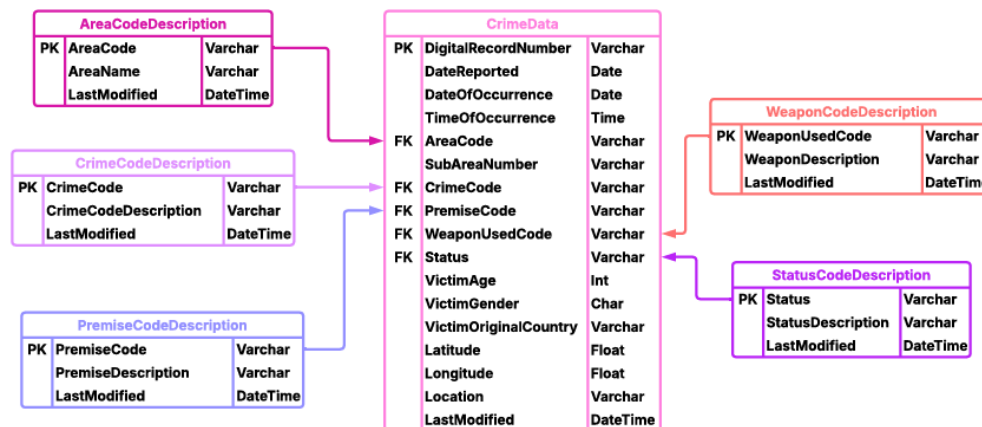
Data Warehousing – May 2025

Data Warehouse Project

Team Members

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Physical Model of Crime Database



Define the Business Processes that you will model

The data warehouse will model several critical business processes related to crime data management and analysis. These processes are essential for supporting decision-making, resource allocation, and public safety efforts. The key business processes include:

1. **Crime Reporting**

Capturing and analyzing data about crimes reported by citizens or law enforcement officers, including when and where the crimes occurred.

2. Crime Occurrence Tracking

Monitoring the date, time, and location of crime incidents to identify patterns and high-crime areas.

3. Crime Classification

Analyzing the types of crimes committed using standardized crime codes and descriptions.

4. Victim Demographics Monitoring

Recording and analyzing information about victims such as age, gender, and nationality to understand vulnerable population groups.

5. Weapon Usage Analysis

Identifying the types of weapons used in criminal activities and their frequency.

6. Case Status Tracking

Monitoring the status of cases (e.g., open, closed, pending) to evaluate resolution rates and workload.

7. Area-Based Crime Analysis

Comparing crime statistics across different areas and sub-areas to support local-level intervention and policing strategies.

8. Trend Analysis Over Time

Observing crime trends over time (e.g., daily, monthly, yearly) to detect spikes or declines in certain types of crime.

Declare the grain of each fact table:

1. FactCrimeSummary

- **Grain:** One row per combination of **Date, Area, Crime Type, Weapon, and Premise**.
 - **Justification:** The table stores aggregated crime counts (CrimeCount) at the intersection of these dimensions.
-

2. FactVictimAnalysis

- **Grain:** One row per **Crime Type and Date**.
 - **Justification:** The table includes victim statistics (count, sum of ages, average age) per crime type on a specific day — no individual-level victim data.
-

3. FactCrimeTime

- **Grain:** One row per combination of **Date, Time, and Area**.
 - **Justification:** It aggregates the number of crimes that occurred in each area during specific time intervals on a given date.
-

4. FactCaseStatus

- **Grain:** One row per **Status, Area, and Date**.
 - **Justification:** Captures the count of cases by status (e.g., "Adult Arrest", "Juv Other") for each area and date.
-

Define the type of each fact table:

All fact tables in this data warehouse follow the Periodic Snapshot Fact Table design pattern. A Periodic Snapshot captures aggregated measures at regular, predefined intervals (e.g., daily), enabling trend analysis over time. Instead of storing each individual transaction, these fact tables summarize data at specific time points across various dimensions such as date, area, crime type, or status.

This approach is ideal for monitoring historical performance and supporting time-based analysis, such as daily crime counts, victim statistics, and case status trends.

1. **FactCrimeSummary**

- Description: Summarizes crime incidents by date, area, crime type, weapon, and premise.
- Snapshot Interval: Daily.
- Purpose: To reflect on the most recent crime statistics, updated daily.

2. **FactVictimAnalysis**

- Description: Aggregates victim data (count, age) by crime type and date.
 - Snapshot Interval: Daily.
 - Purpose: To capture victim statistics over time, including total victim count and average age.
-

3. **FactCrimeTime**

- Description: Stores crime counts by time of day and area.
 - Snapshot Interval: Daily.
 - Purpose: To provide insights into when and where crimes occur most frequently.
-

4. **FactCaseStatus**

- Description: Tracks case status counts (open, closed, etc.) by area and date.
 - Snapshot Interval: Daily.
 - Purpose: To monitor the progress of criminal cases across different areas and times.
-

Define the dimensions and the type of each one:

1- DimDate

- **Type:** Role-Playing, Conformed
 - **Explanation:**
This dimension is reused in multiple fact tables (e.g., as DateKey, DateReported, DateOfOccurrence), hence it's a role-playing dimension. It is also conformed, meaning it's shared across multiple subject areas (facts) with consistent structure and content.
-

2- DimArea

- **Type:** Slowly Changing Dimension (SCD) Type 1, Conformed
 - **Explanation:**
Changes to area attributes (like name or classification) overwrite previous values. This dimension is conformed, meaning it is used consistently across different fact tables.
-

3- DimCrimeType

- **Type:** Slowly Changing Dimension (SCD) Type 1, Conformed
 - **Explanation:**
Changes in crime type descriptions are overwritten. It is conformed across multiple facts using crime categorization.
-

4- DimTime

- **Type:** Static
 - **Explanation:**
The time dimension (e.g., hour of day, part of day) does not change over time and is considered static. It provides consistent time-of-day groupings for analysis.
-

5- DimWeapon

- **Type:** Slowly Changing Dimension (SCD) Type 1
 - **Explanation:**
Weapon attributes (e.g., classification or naming) are updated by overwriting the old values.
-

6- DimStatus

- **Type:** Slowly Changing Dimension (SCD) Type 1
 - **Explanation:**
Any change in case status (e.g., renaming or reclassification) replaces the previous record, without keeping history.
-

7- DimPremise

- **Type:** Slowly Changing Dimension (SCD) Type 2
 - **Explanation:**
Tracks historical changes in premise attributes by creating a new version of the record with each change. This allows you to preserve history and analyze how premises changed over time.
-

Define the measures that will appear in the fact tables and the type of each one

FactCrimeSummary

1. CrimeCount

- **Type:** Basic Measure
 - **Aggregation Type:** Additive
 - **Explanation:** Can be summed across all dimensions (e.g., date, area, crime type).
-

FactVictimAnalysis

1. VictimCount

- **Type:** Basic Measure
- **Aggregation Type:** Additive
- **Explanation:** Total number of victims, summable over time or categories.

2. SumVictimAge

- **Type:** Basic Measure
- **Aggregation Type:** Additive
- **Explanation:** Total age of all victims can be summed across dimensions.

3. AvgVictimAge

- **Type:** Calculated Measure
- **Aggregation Type:** Treated as Additive (because it is calculated from two additive measures)
- **Explanation:**
 - Derived as: $\text{AvgVictimAge} = \text{SumVictimAge} / \text{VictimCount}$.

FactCrimeTime

1. CrimeCount

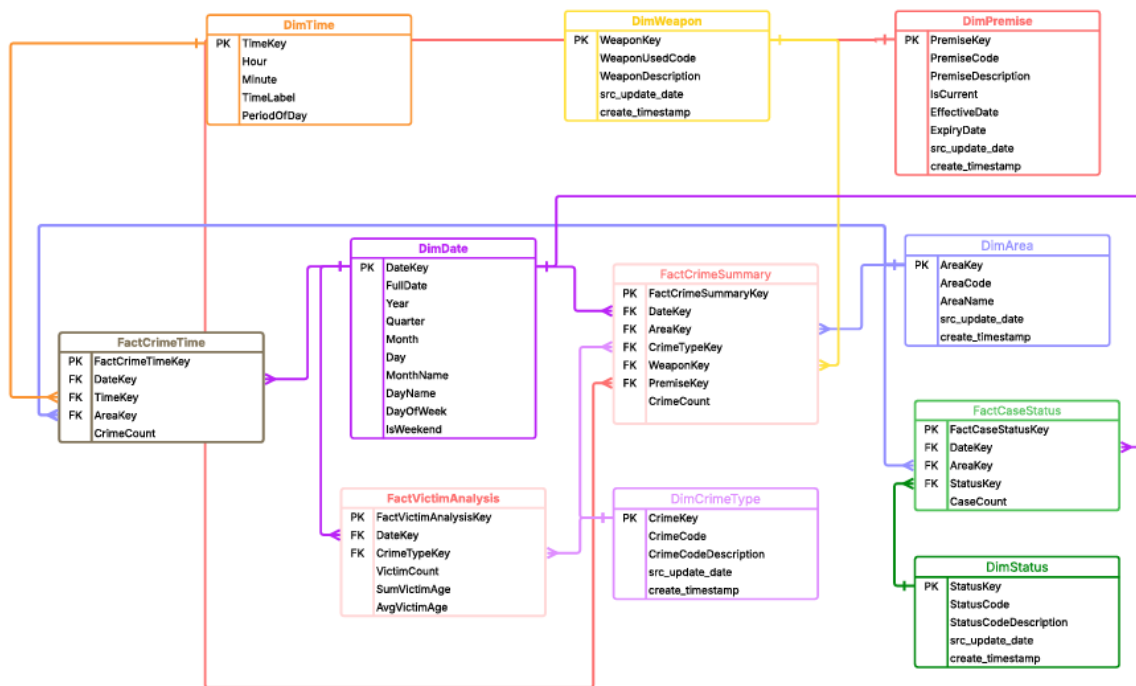
- **Type:** Basic Measure
 - **Aggregation Type:** Additive
 - **Explanation:** Represents number of crimes by time and area, summable across dimensions.
-

FactCaseStatus

1. CaseCount

- **Type:** Basic Measure
- **Aggregation Type:** Additive
- **Explanation:** Tracks number of cases by status, date, and area; summable across those.

Physical model of the Crime Data Warehouse(Star Schema)

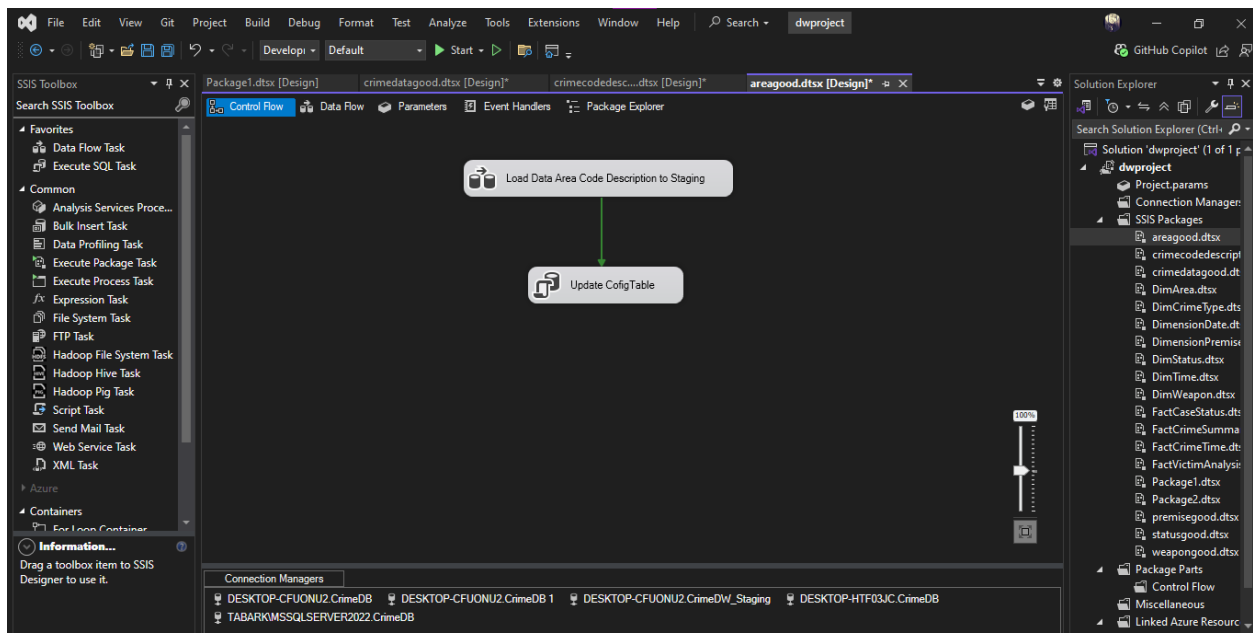


Screenshots of the data flow tasks, and control flow tasks used for building the DWH

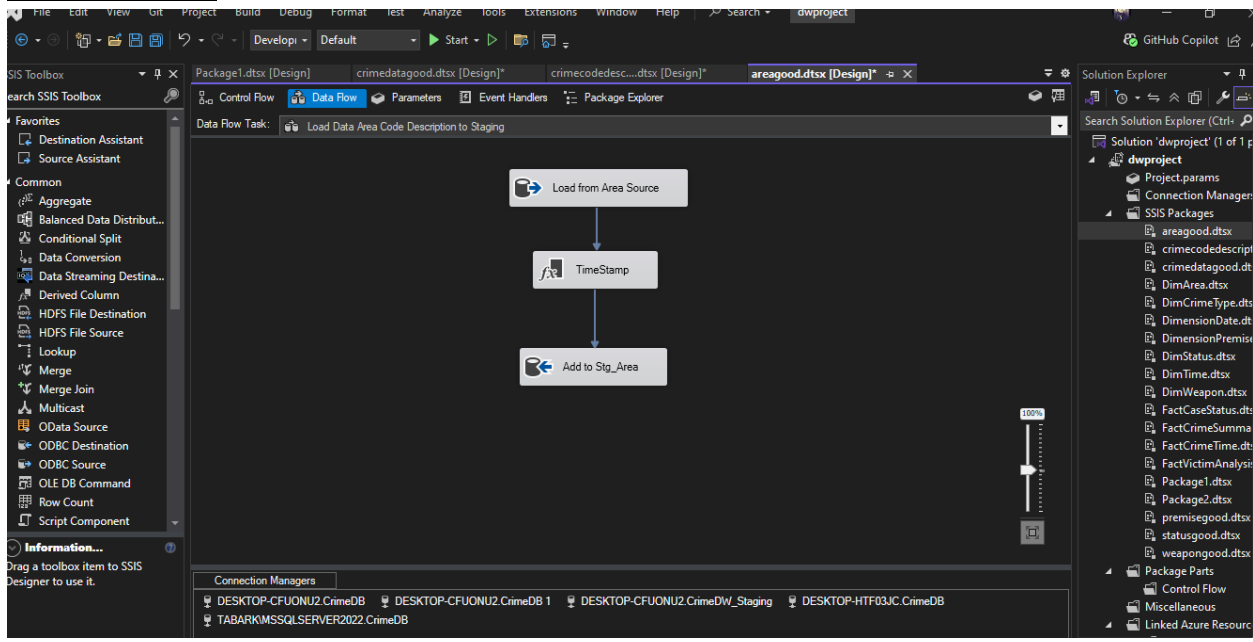
Stagging Tables

From Source (AreaCodeDescription) to (Stg-Area)

Control Flow

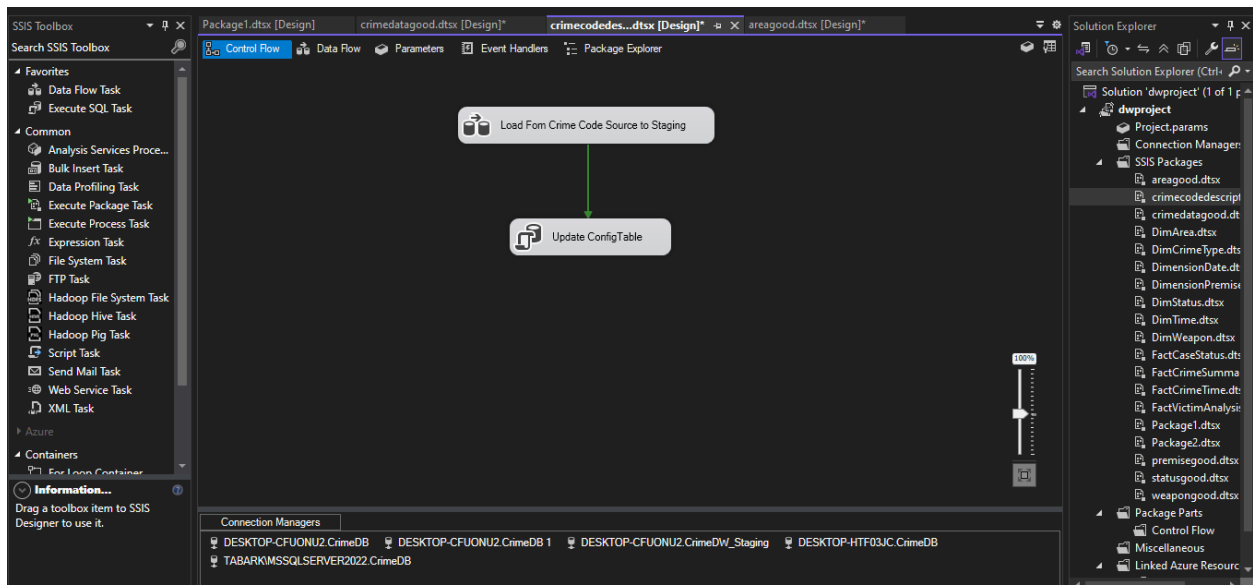


Data Flow

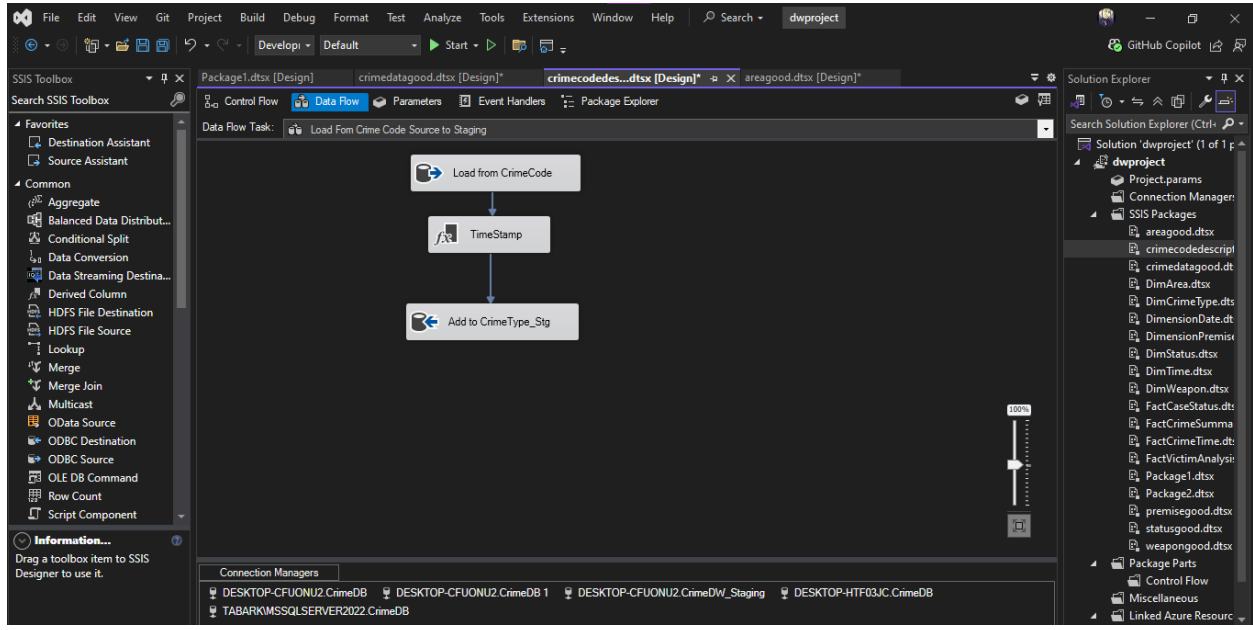


From Source (CrimeCodeDescription) to (Stg-CrimeType)

Control Flow

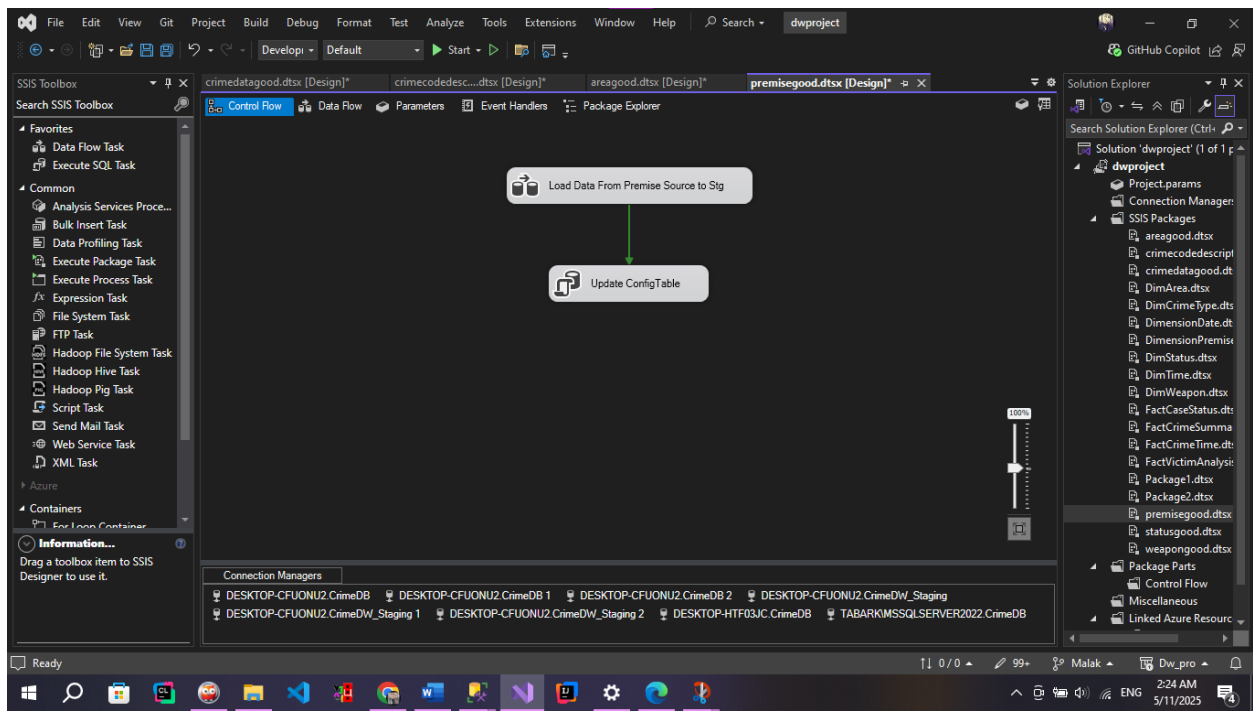


Data Flow

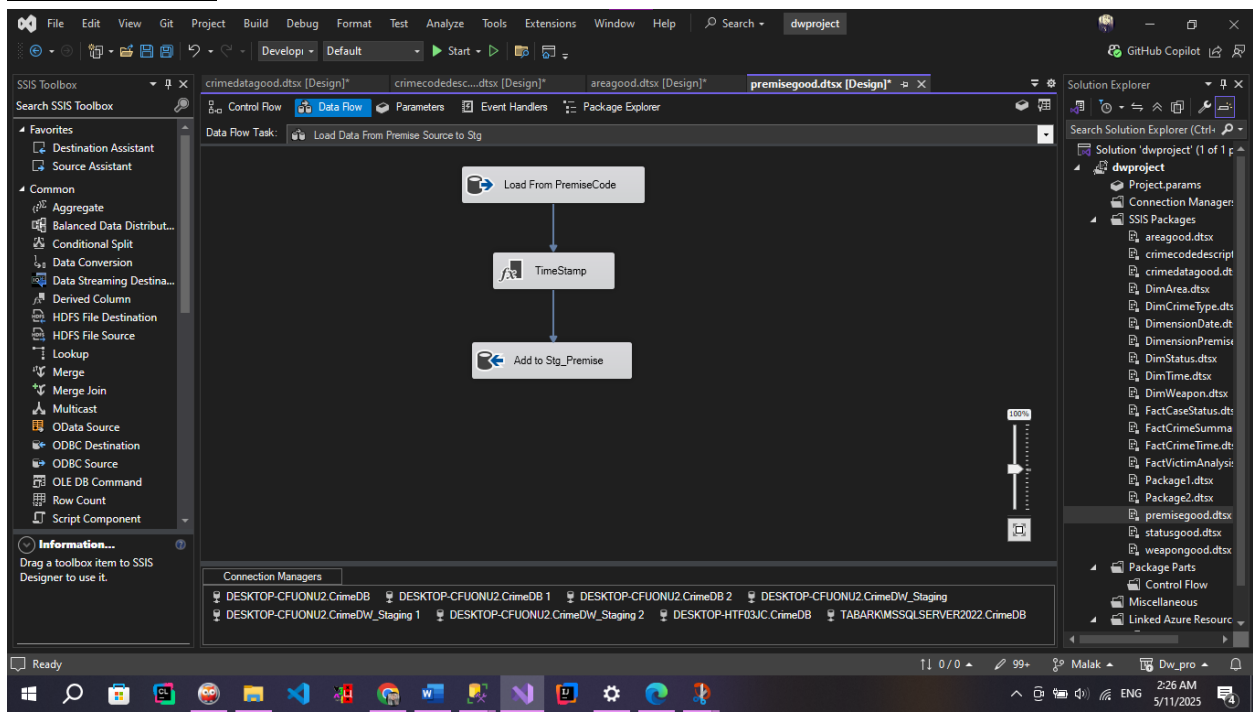


From Source (PremiseCodeDescription) to (Stg-Premise)

Control Flow

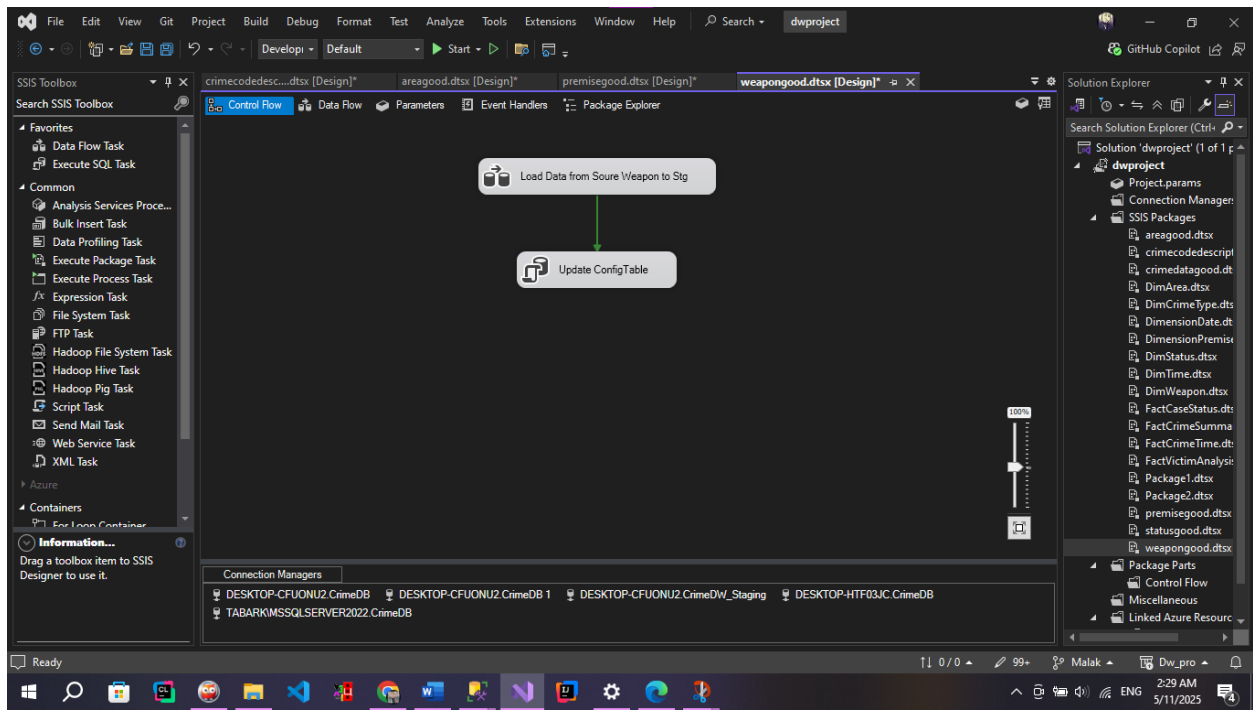


Data Flow

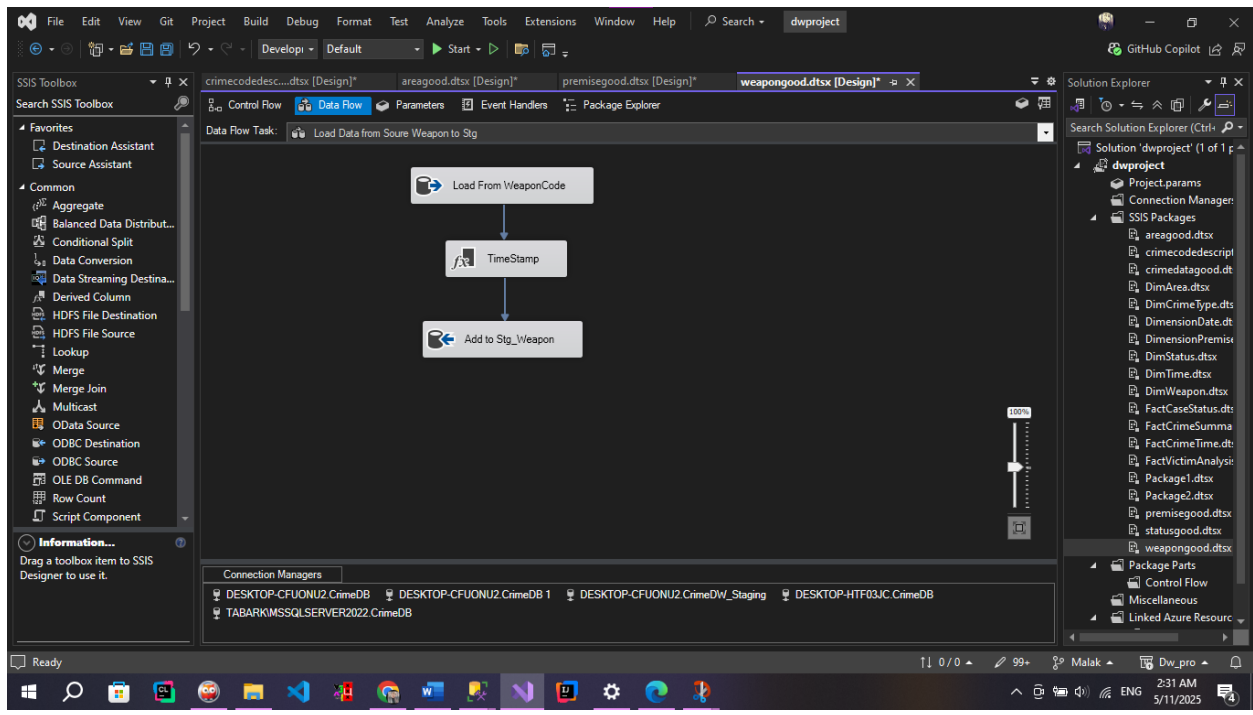


From Source (WeaponCodeDescription) to (Stg-Weapon)

Control Flow

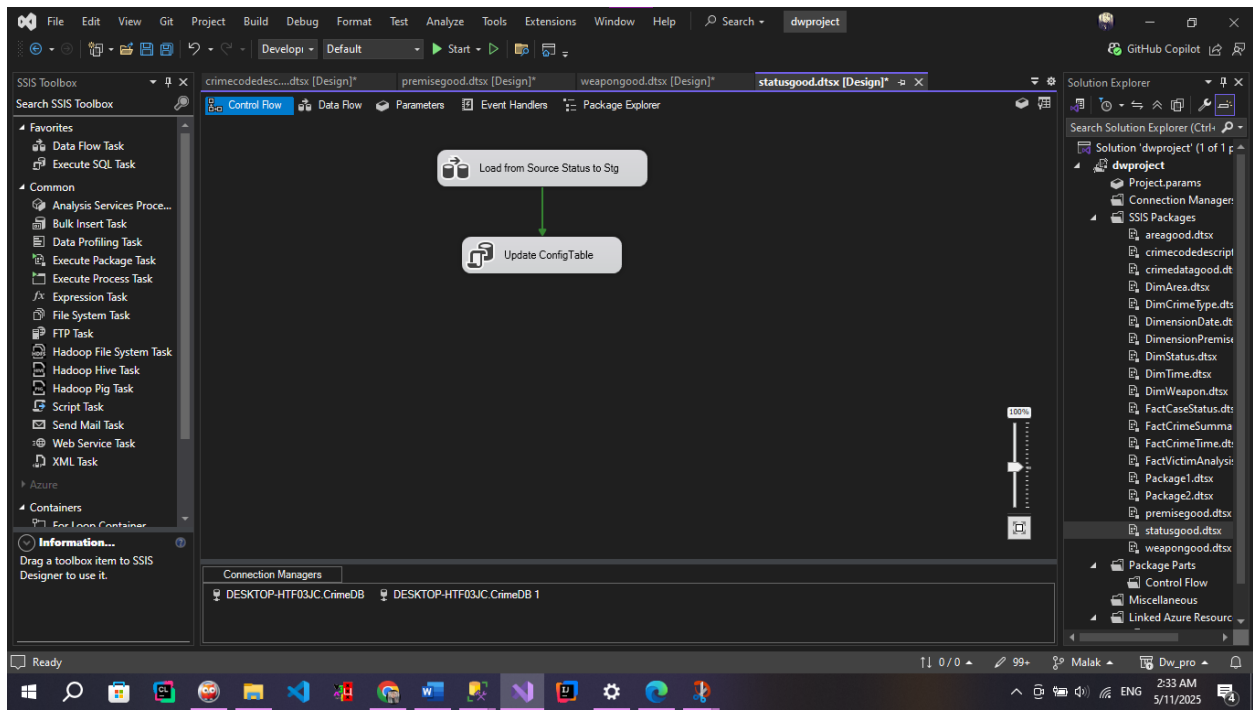


Data Flow

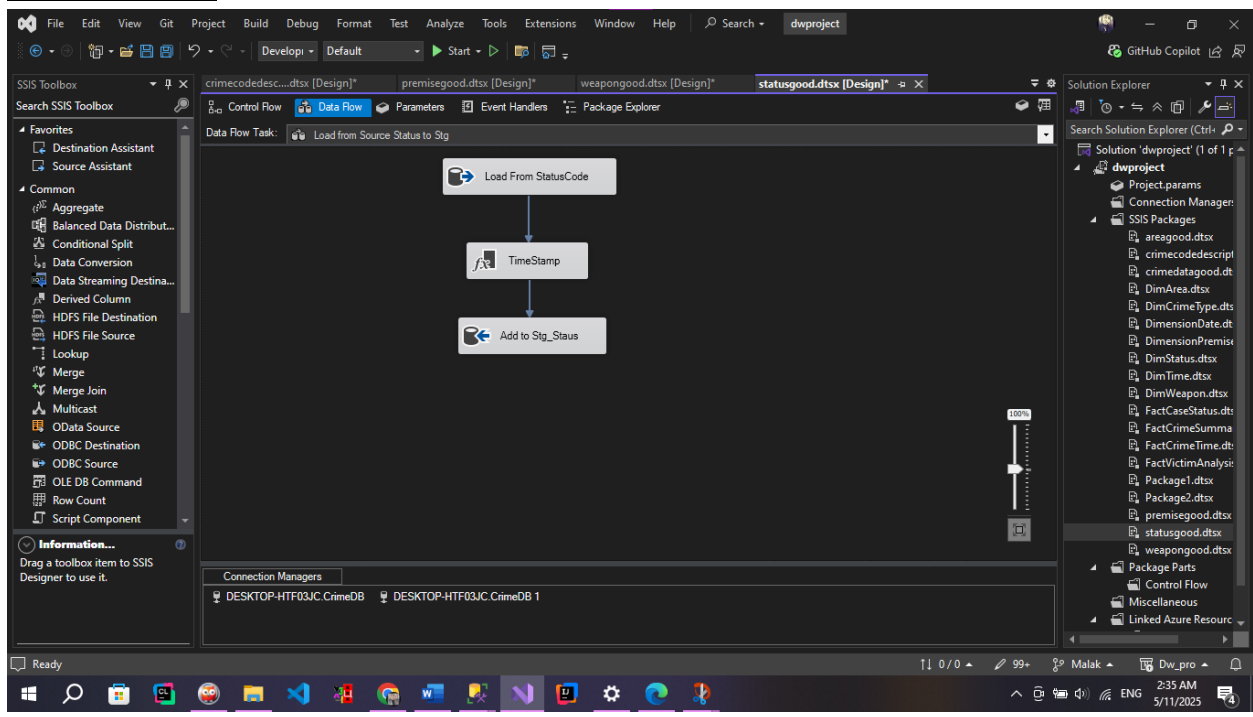


From Source (StatusCodeDescription) to (Stg-Status)

Control Flow

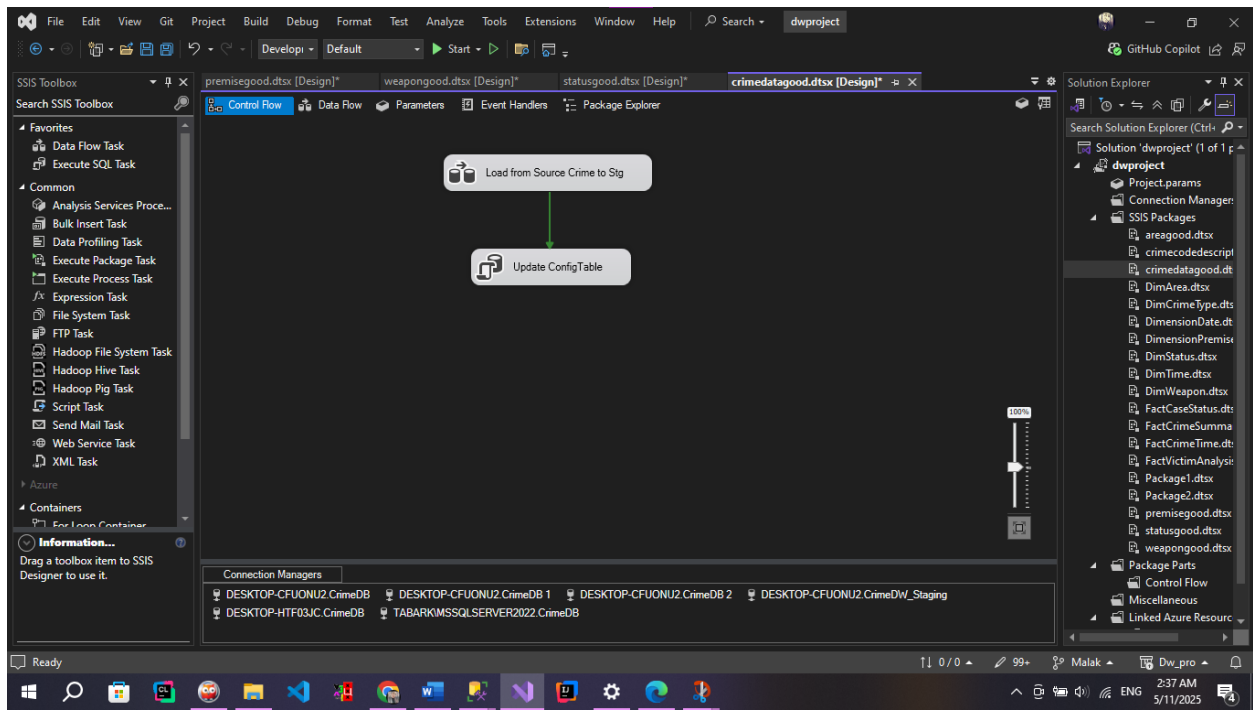


Data Flow

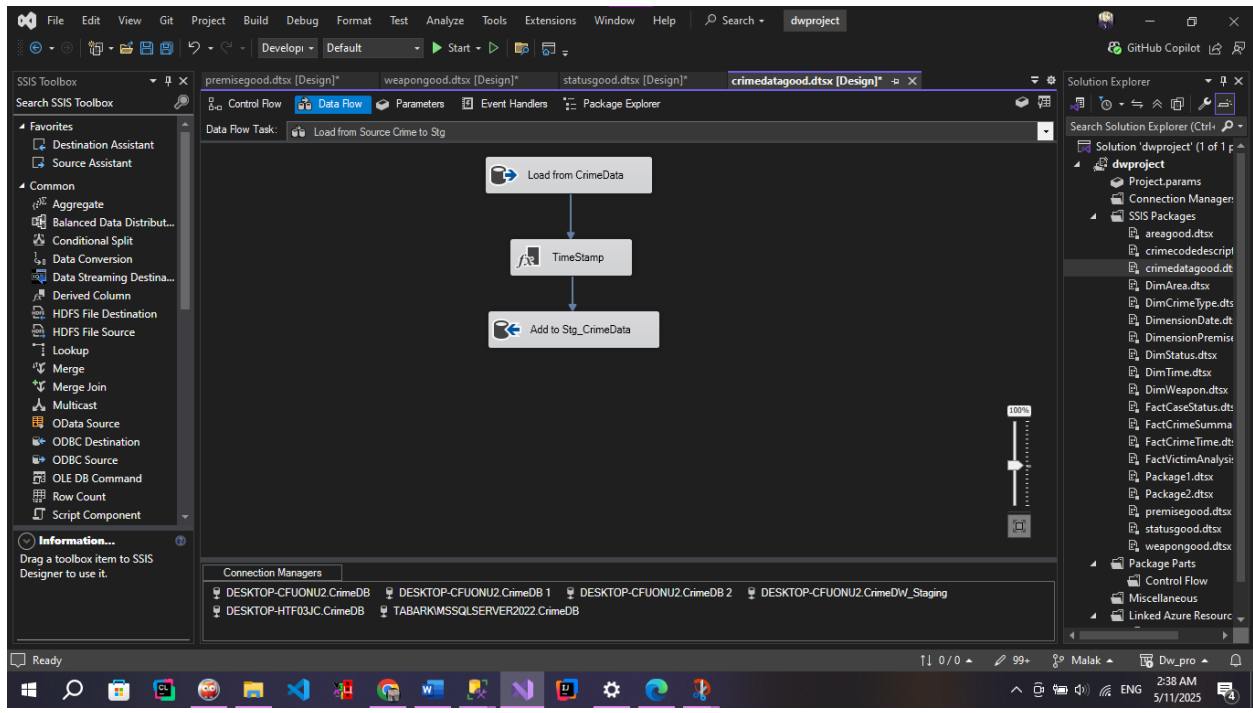


From Source (CrimeData) to (Stg-CrimeData)

Control Flow

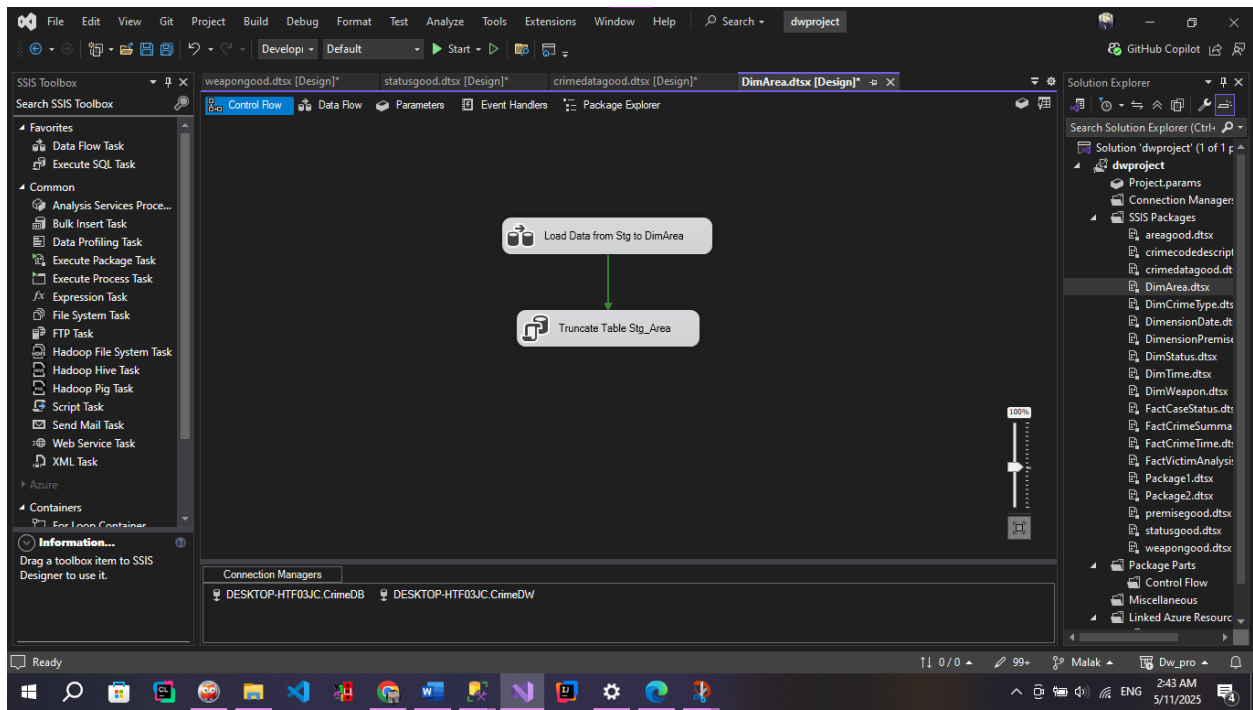


Data Flow

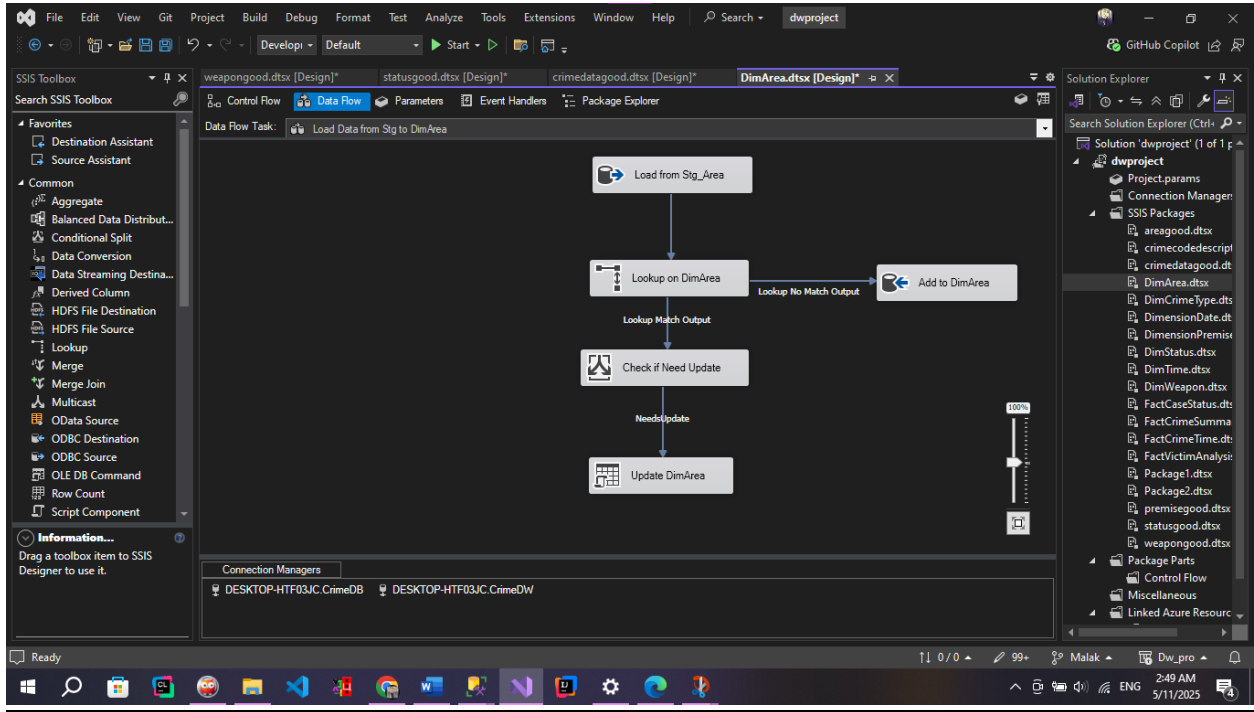


Dimensions

Load From (Stg-Area) to (DimArea) Control Flow

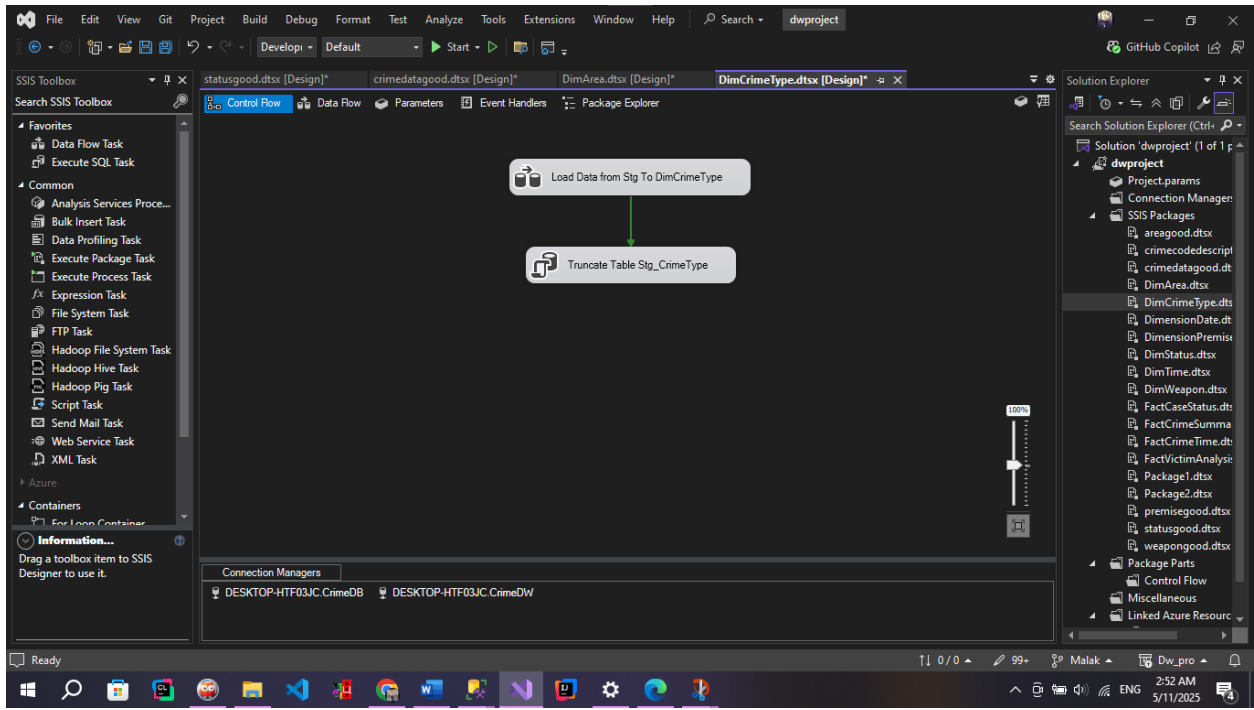


Data Flow

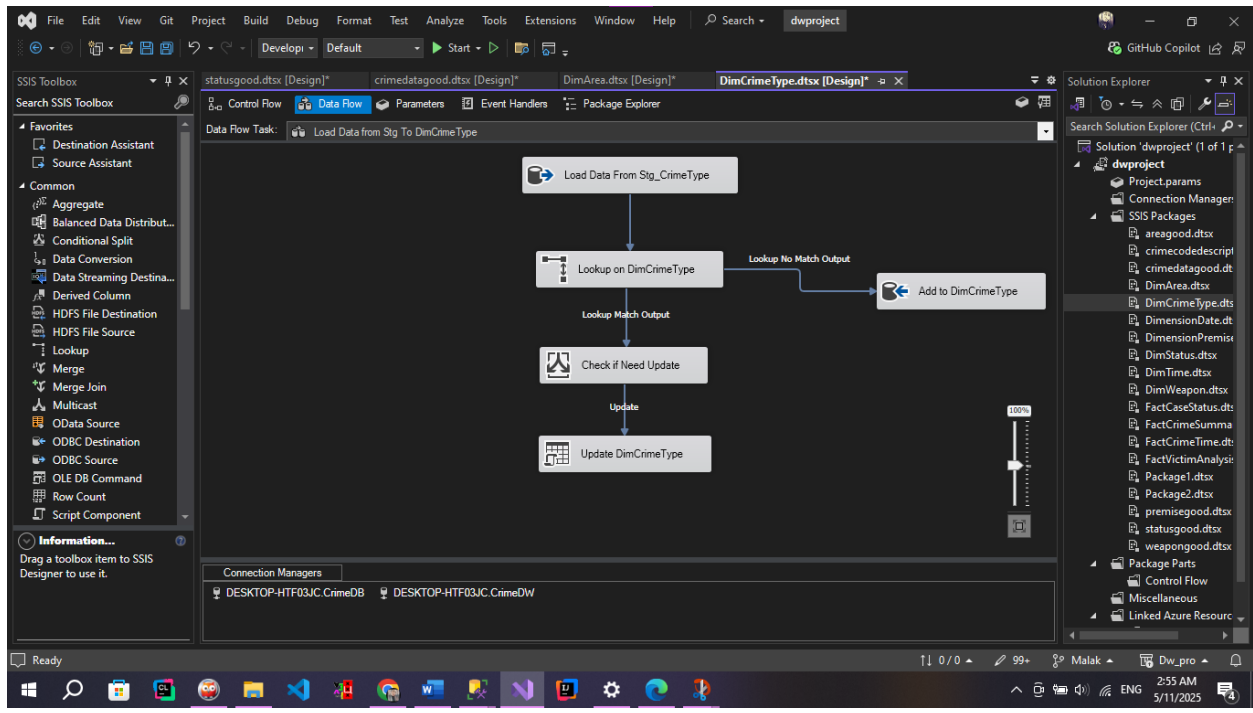


Load From (Stg-CrimeType) to (DimCrimeType)

Control Flow

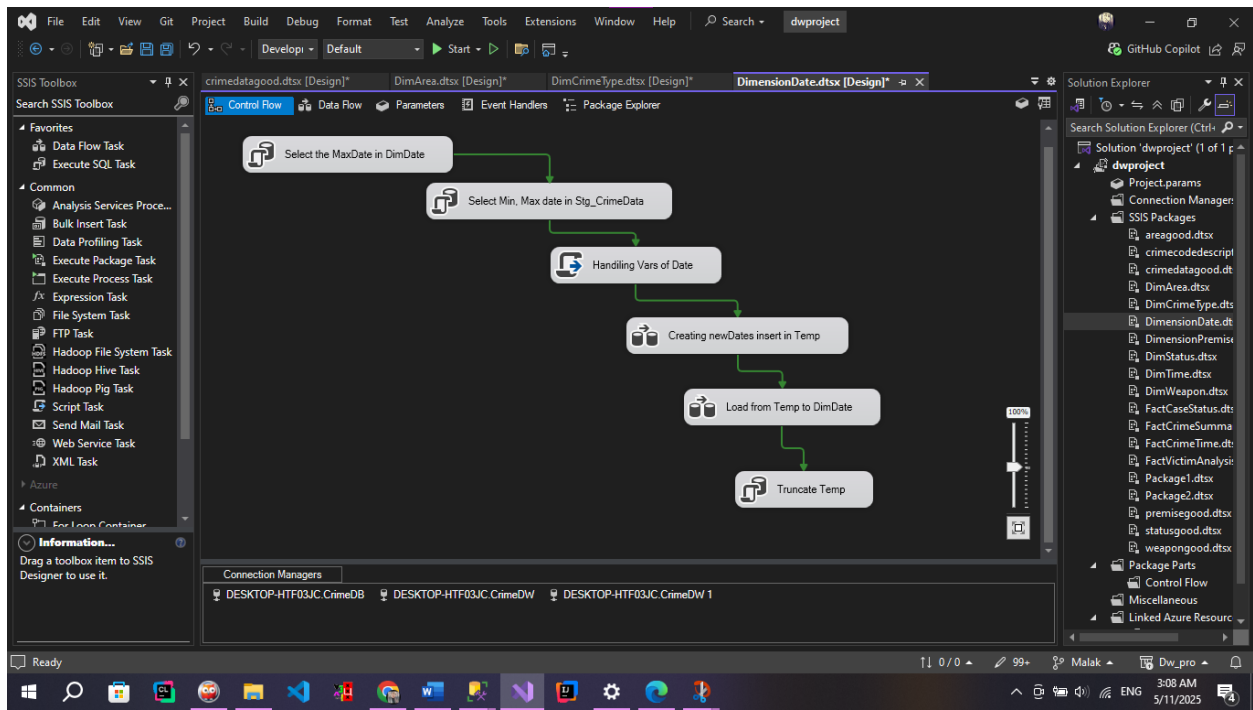


Data Flow

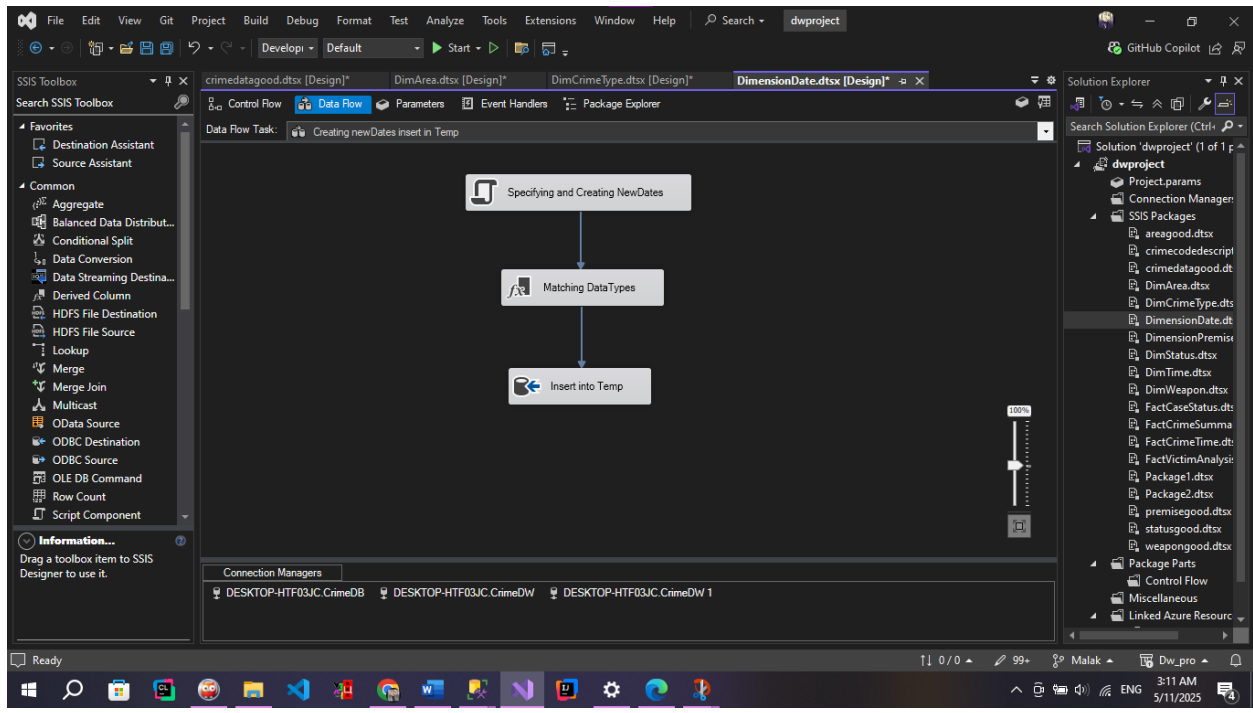


Load From (Stg-CrimeData) to (DimDate)

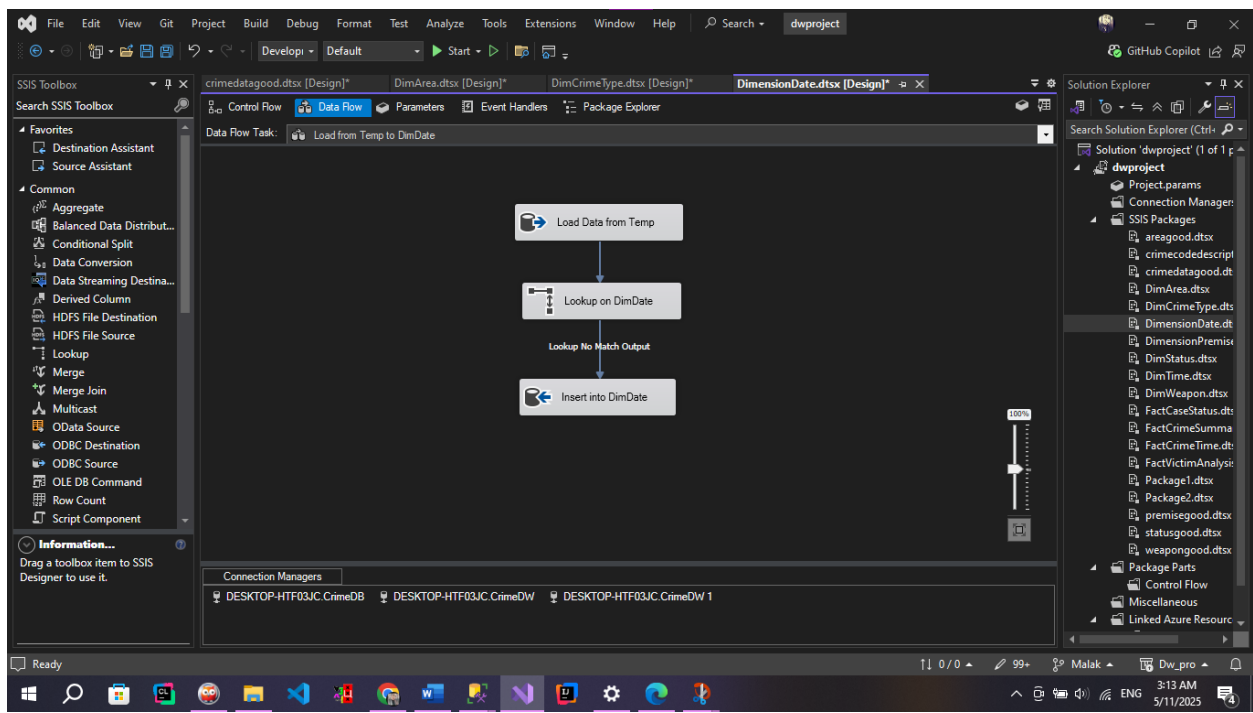
Control Flow



Data Flow(Creating newDates insert in Temp)

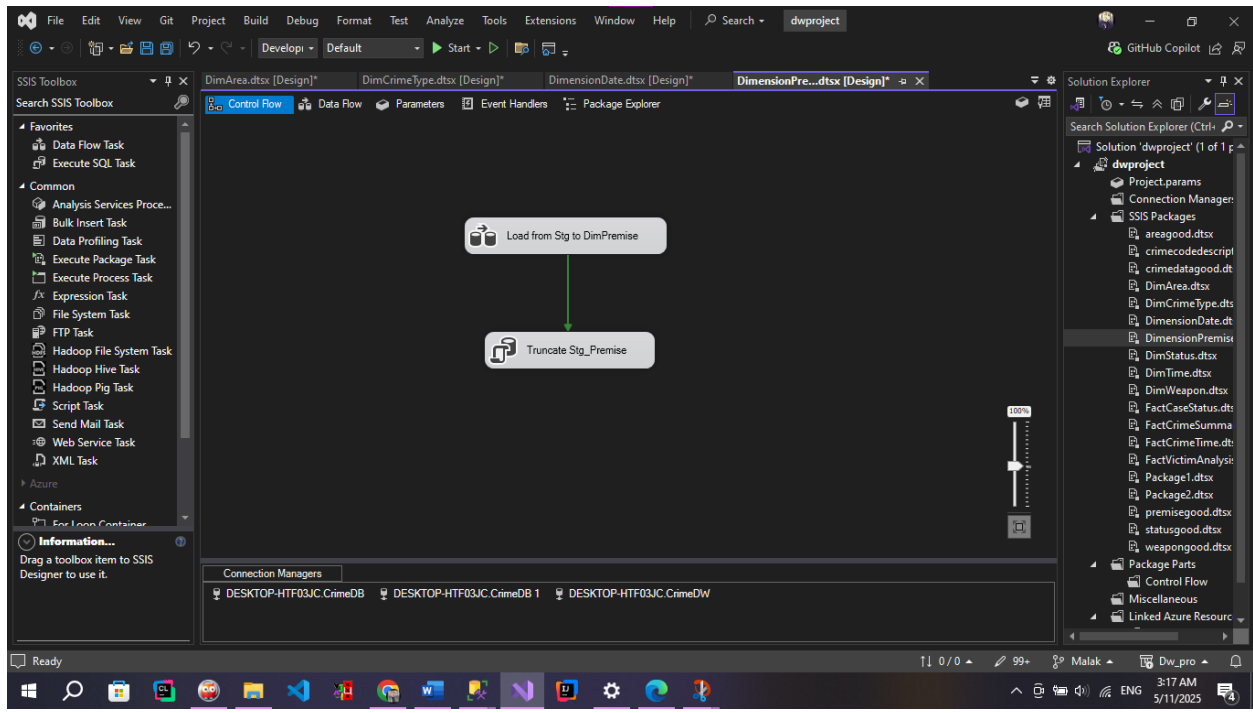


Data Flow (Load from Temp to DimDate)

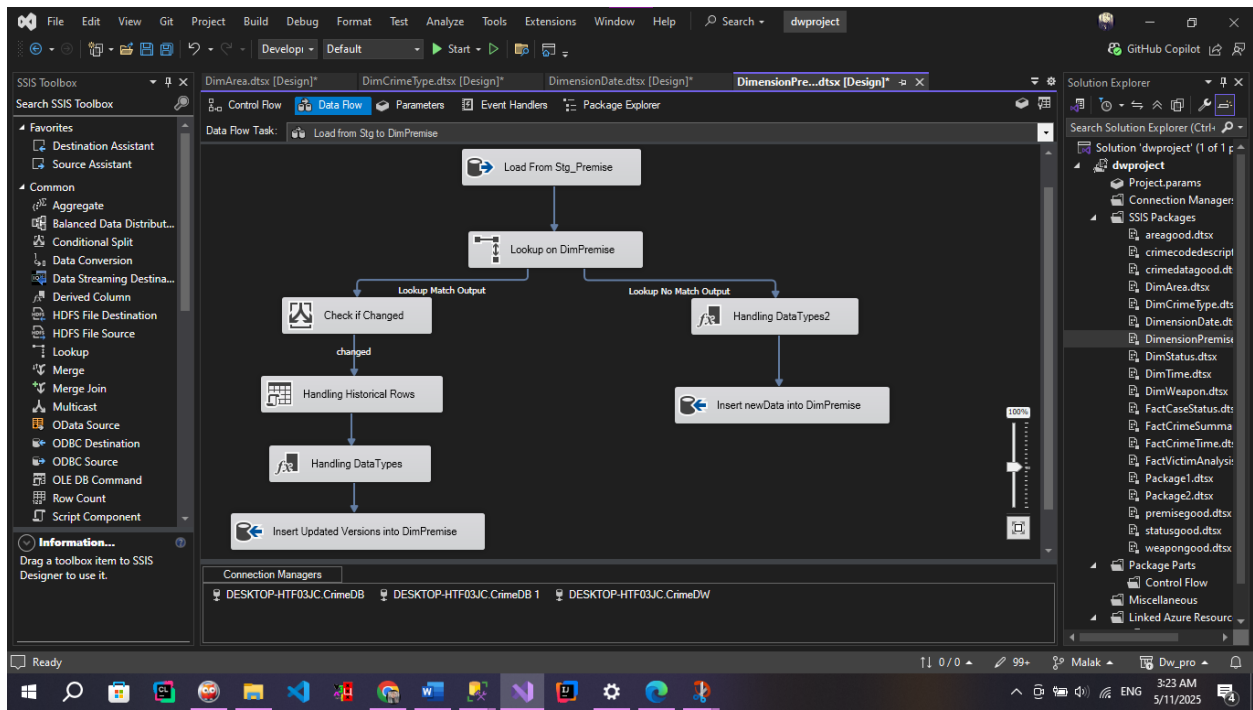


Load From (Stg-Premise) to (DimPremise)

Control Flow

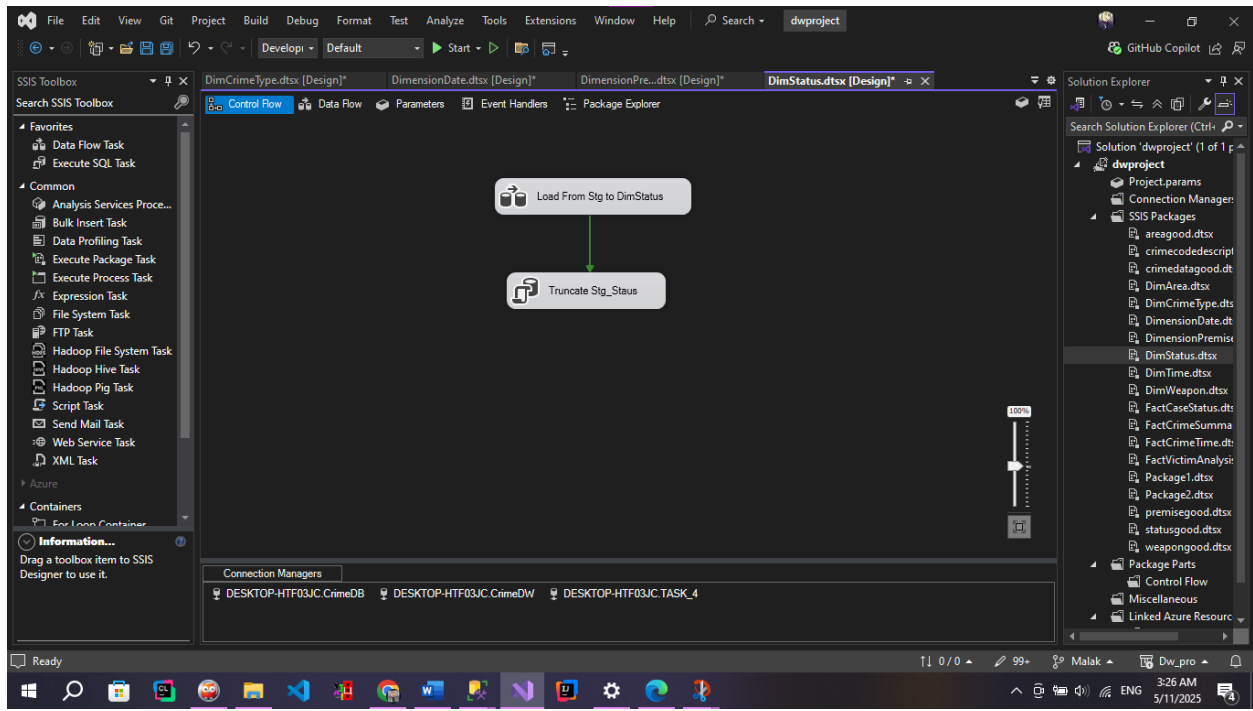


Data Flow

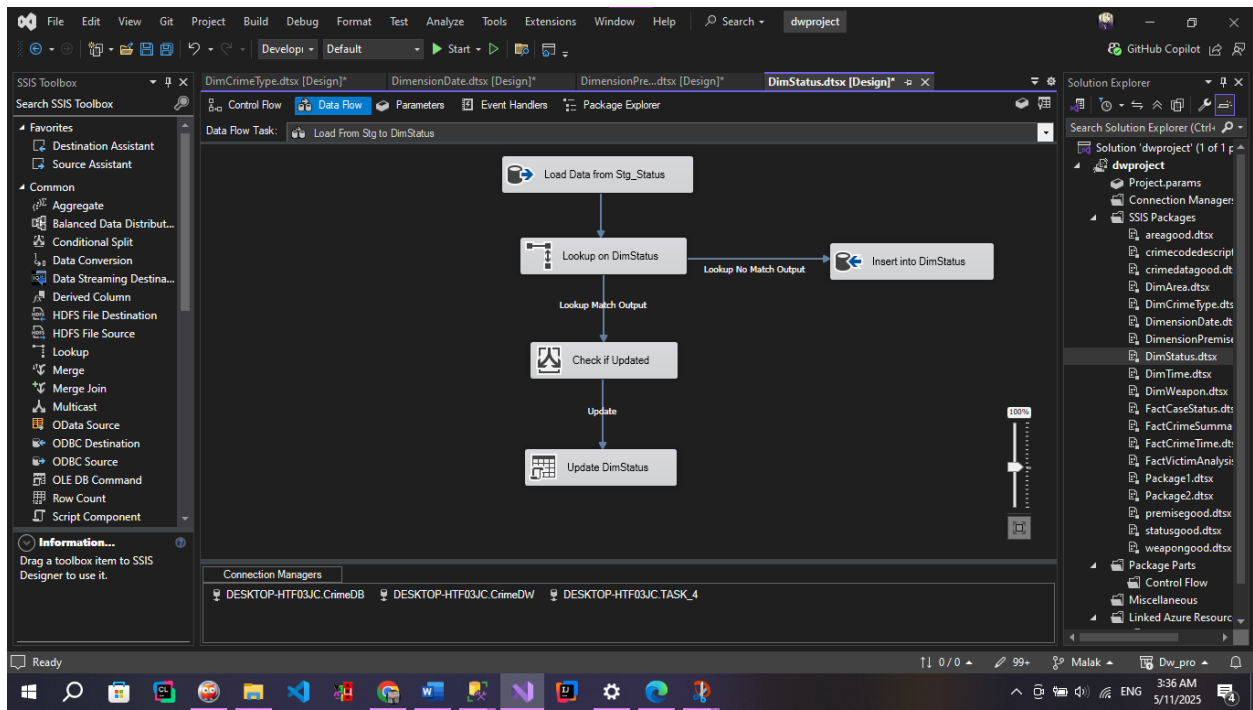


Load From (Stg-Status) to (DimStatus)

Control Flow

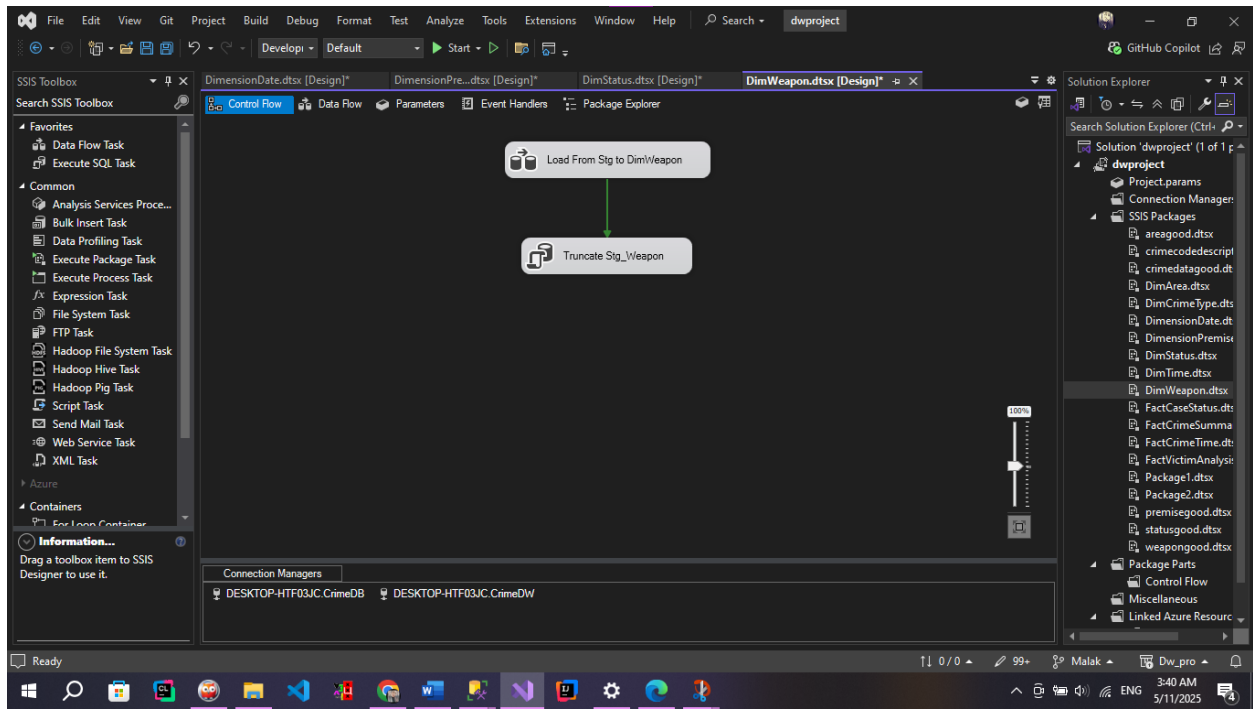


Data Flow

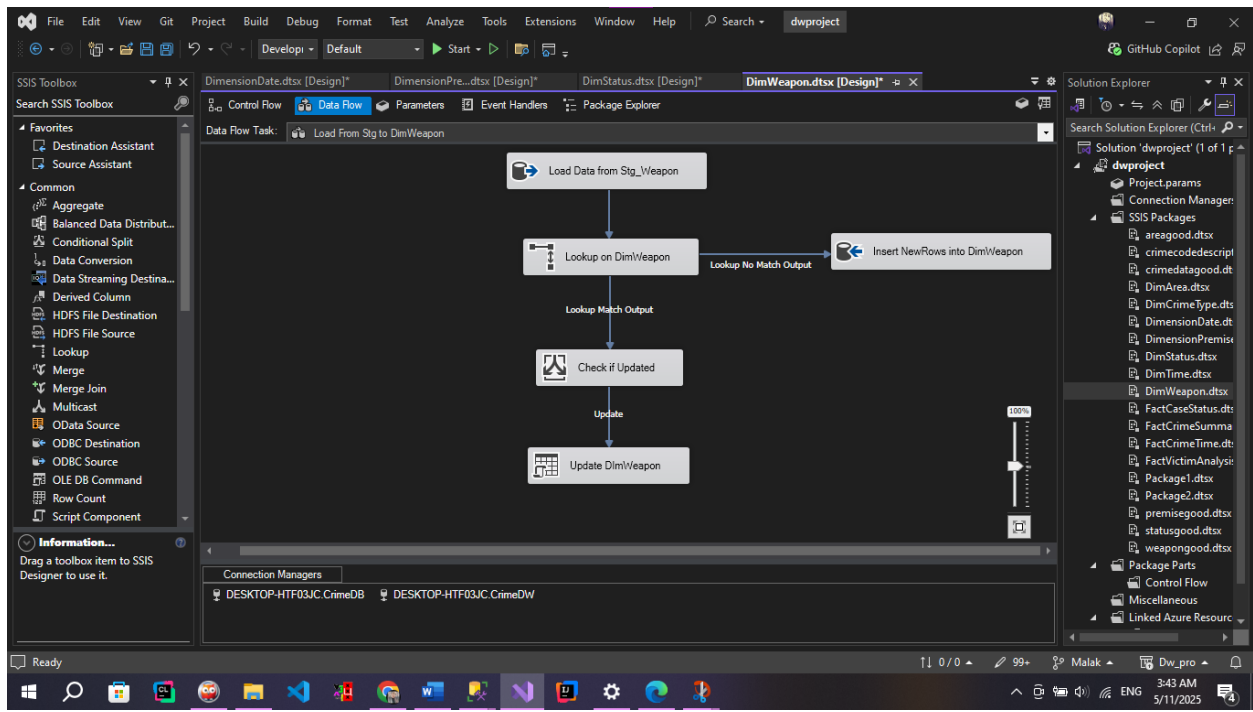


Load From (Stg-Weapon) to (DimWeapon)

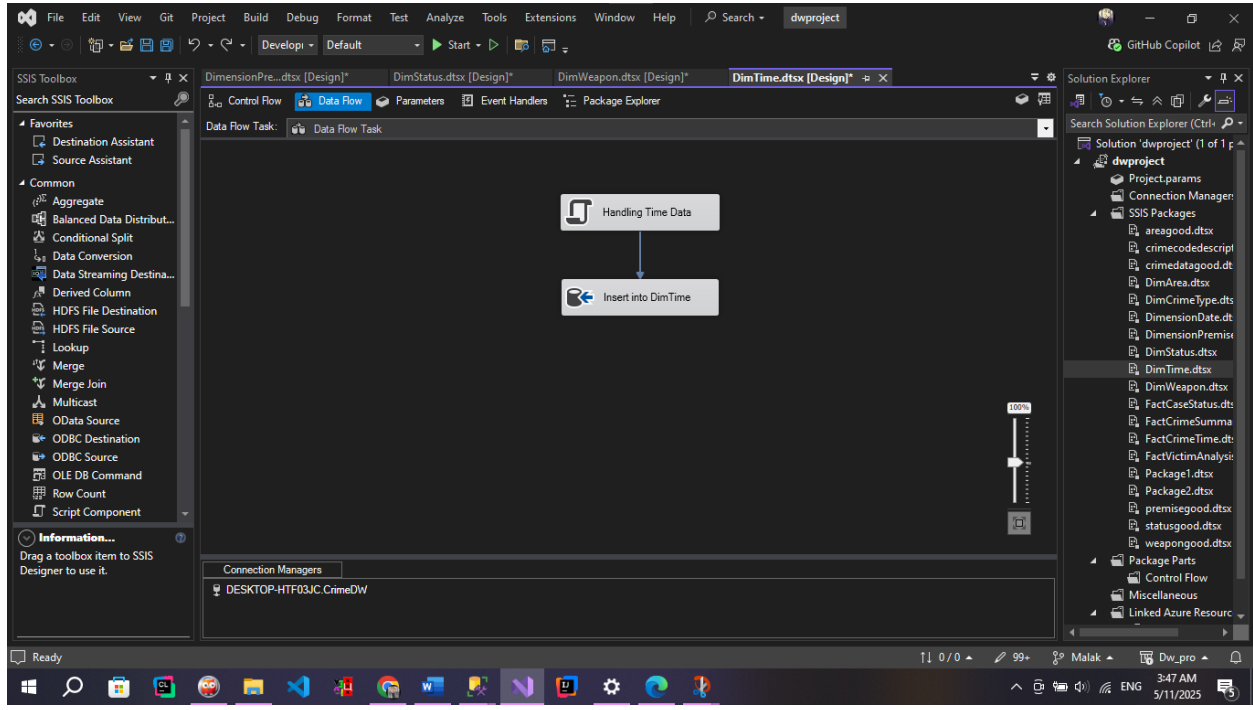
Control Flow



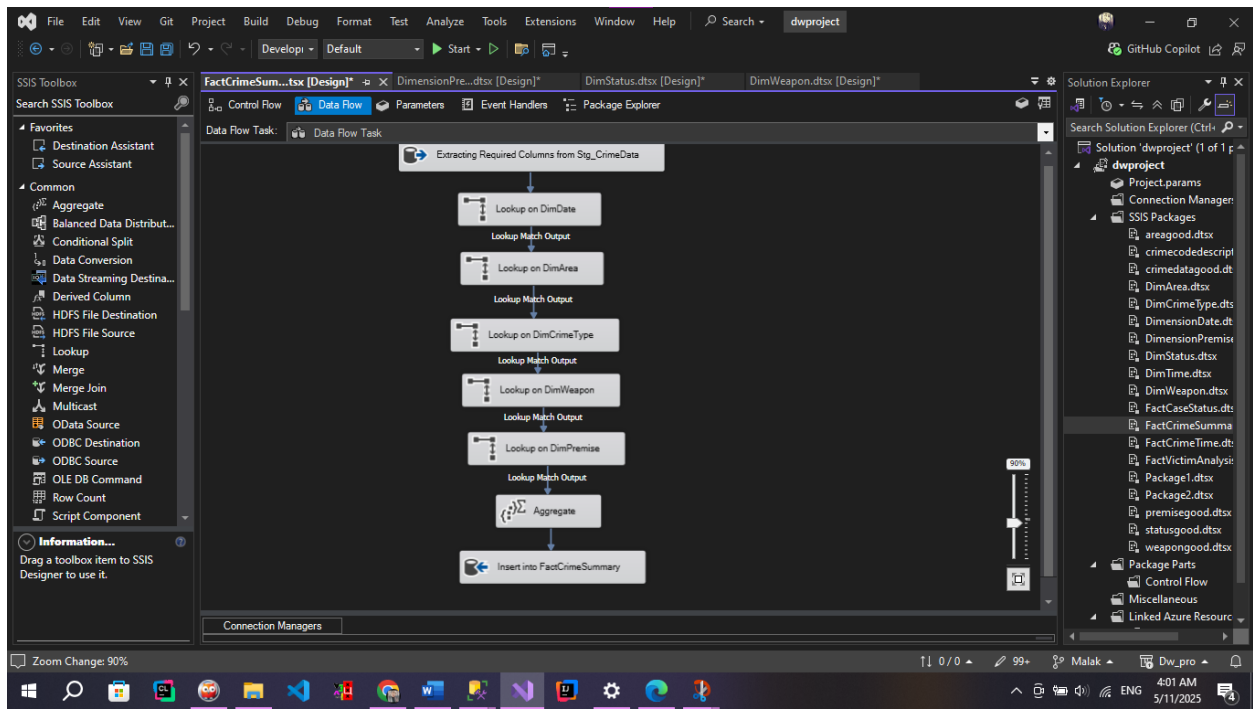
Data Flow



Insert into (DimTime) Data Flow

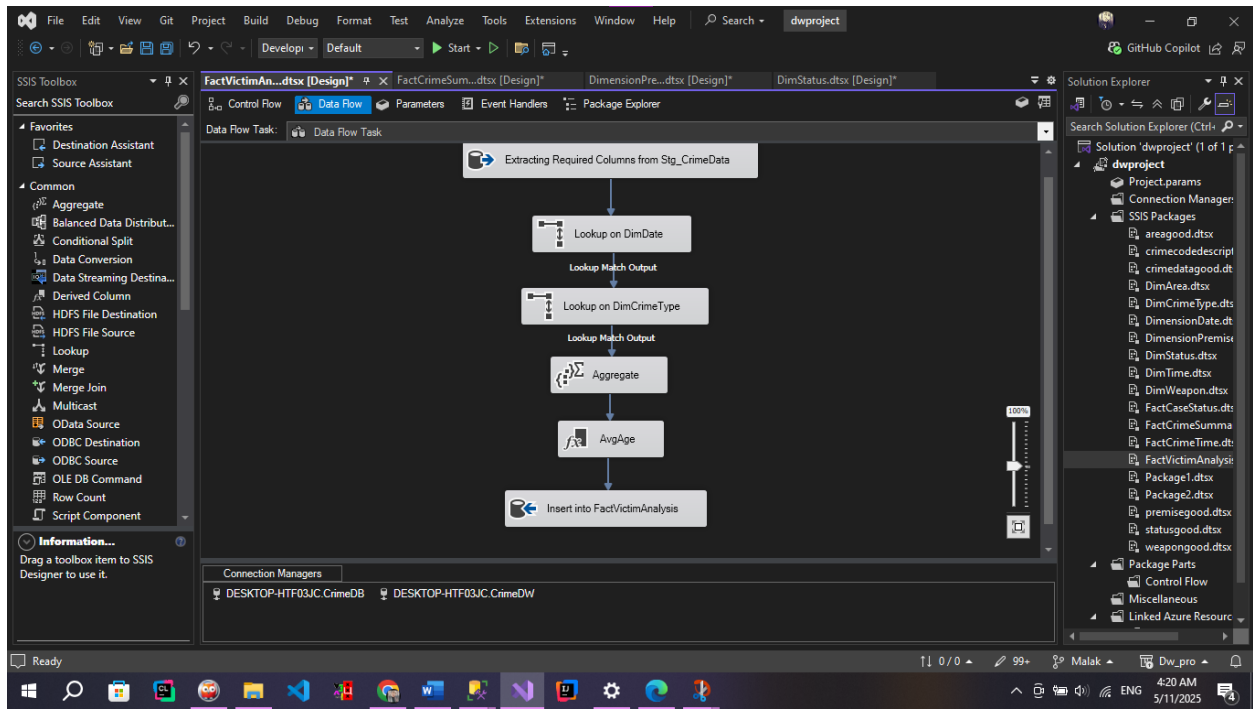


Fact Tables Load From Dim(Date, Area, Premise, CrimeType, Premise) to FactCrimeSummary Data Flow

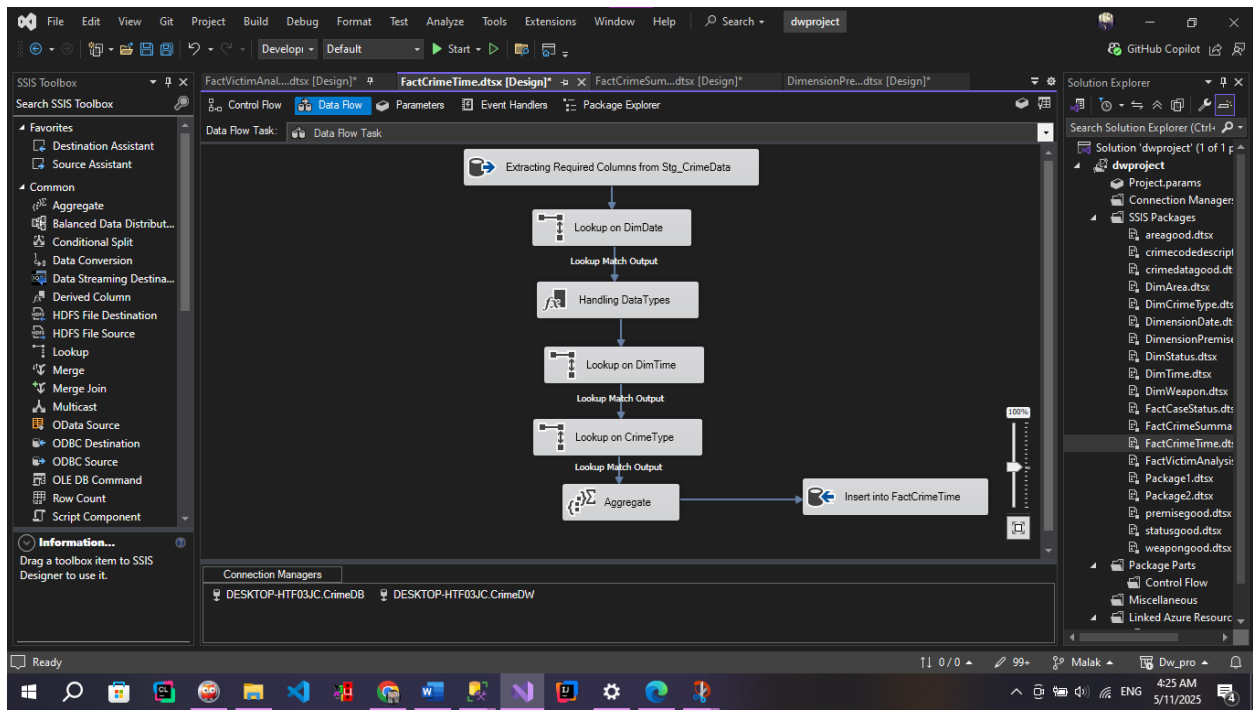


Load From Dim(Date, CrimeType) into FactVictimAnalysis

DataFlow

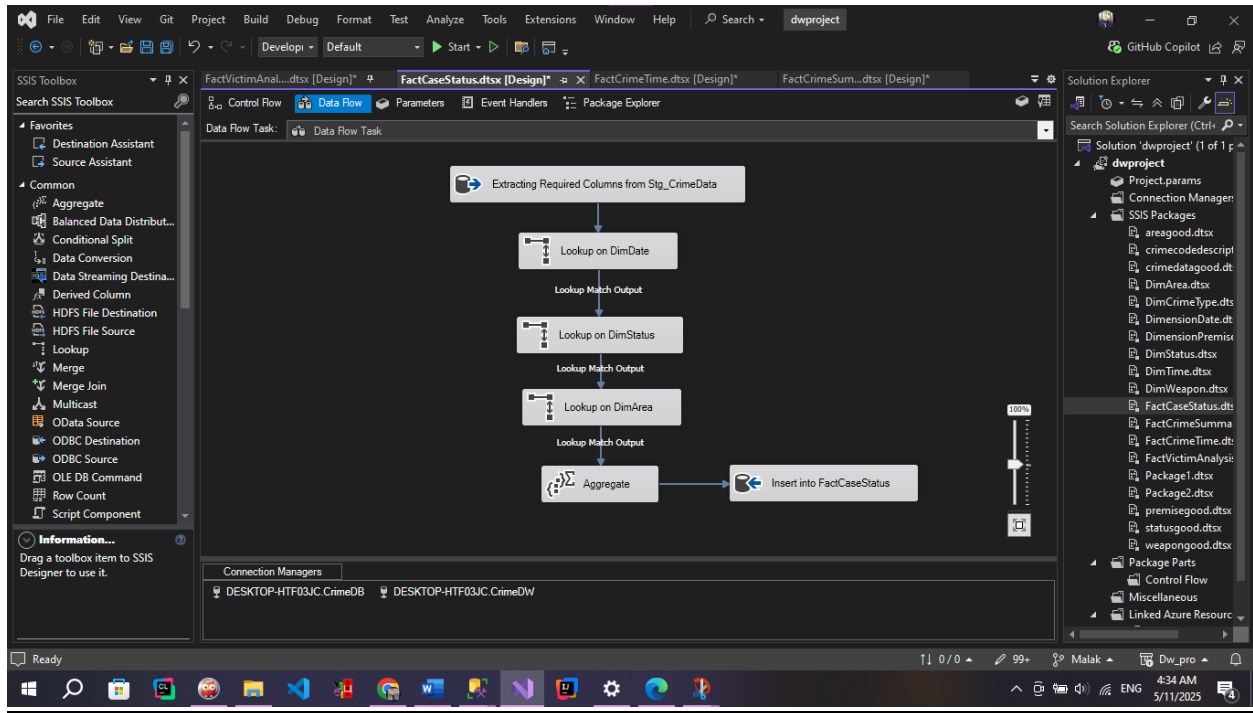


Load From Dim(Date, Time, Area) into FactCrimeTime DataFlow



Load From Dim(Date, Status, Area) into FactCaseStatus

DataFlow



Queries (FactCrimeSummary)

```
-- 1. Total number of crimes per area
SELECT
    a.AreaCode      AS [Area Code],
    a.AreaName      AS [Area Name],
    SUM(f.CrimeCount) AS TotalCrimes
FROM FactCrimeSummary f
JOIN DimArea a ON f.AreaKey = a.AreaKey
GROUP BY
    a.AreaCode,
    a.AreaName
ORDER BY
    TotalCrimes DESC;
```

100 % 1. Total number of crimes per area

Results Messages

	Area Code	Area Name	TotalCrimes
1	1	Central	37675
2	14	Pacific	35603
3	12	77th Street	32640
4	15	N Hollywood	30054
5	7	Wilshire	29875
6	3	Southwest	29482
7	13	Newton	28491
8	8	West LA	27739
9	6	Hollywood	27389
10	11	Northeast	27358
11	17	Devonshire	26666
12	18	Southeast	26647
13	9	Van Nuys	25250
14	20	Olympic	25226
15	10	West Valley	24887
16	21	Topanga	24770
17	2	Rampart	23257
18	19	Mission	23239
19	5	Harbor	23061
20	4	Hollenbeck	22244
21	16	Foothill	20045

```
-- 2. Most frequent crime types
```

```
SELECT
    ct.CrimeCode      AS [Crime Code],
    ct.CrimeCodeDescription AS [Crime Type],
    SUM(f.CrimeCount) AS Occurrences
FROM FactCrimeSummary f
JOIN DimCrimeType ct ON f.CrimeTypeKey = ct.CrimeKey
GROUP BY
    ct.CrimeCode,
    ct.CrimeCodeDescription
ORDER BY
    Occurrences DESC;
```

	Crime Code	Crime Type	Occurrences
1	510	Vehicle - Stolen	115102
2	330	Burglary From Vehicle	60184
3	310	Burglary	47848
4	354	Theft Of Identity	45484
5	420	Theft From Motor Vehicle - Petty (\$950 & Under)	40066
6	440	Theft Plain - Petty (\$950 & Under)	39729
7	331	Theft From Motor Vehicle - Grand (\$950.01 An...	35044
8	442	Shoplifting - Petty Theft (\$950 & Under)	29081
9	210	Robbery	17341
10	888	Trespassing	15661
11	745	Vandalism - Misdemeanor (\$399 Or Under)	14712
12	930	Criminal Threats - No Weapon Displayed	14594
13	761	Brandish Weapon	11718
14	901	Violation Of Restraining Order	8555
15	480	Bike - Stolen	6735
16	624	Battery - Simple Assault	6173
17	946	Other Miscellaneous Crime	5851
18	343	Shoplifting-Grand Theft (\$950.01 & Over)	5490
19	900	Violation Of Court Order	4638
20	520	Vehicle - Attempt Stolen	3716
21	352	Pickpocket	3618
22	236	Intimate Partner - Aggravated Assault	3535
23	649	Document Forgery / Stolen Felony	2854
24	220	Attempted Robbery	2711
25	753	Discharge Firearms/Shots Fired	2562

-- 3. Most used weapon types

```
SELECT
    w.WeaponUsedCode      AS [Weapon Code],
    w.WeaponDescription   AS [Weapon Name],
    SUM(f.CrimeCount) AS TimesUsed
FROM FactCrimeSummary f
JOIN DimWeapon w ON f.WeaponKey = w.WeaponKey
GROUP BY
    w.WeaponUsedCode,
    w.WeaponDescription
ORDER BY
    TimesUsed DESC;
```

Results

Messages

	Weapon Code	Weapon Name	TimesUsed
1	0	Unknown	488653
2	500	Unknown Weapon/Other Weapon	22692
3	511	Verbal Threat	18260
4	102	Hand Gun	11036
5	109	Semi-Automatic Pistol	3719
6	200	Knife With Blade 6Inches Or Less	3381
7	207	Other Knife	2892
8	106	Unknown Firearm	2880
9	113	Simulated Gun	1163
10	306	Rock/Thrown Object	1142
11	204	Folding Knife	1141
12	308	Stick	838
13	512	Mace/Pepper Spray	821
14	201	Knife With Blade Over 6 Inches L...	772
15	307	Vehicle	764
16	515	Physical Presence	739
17	205	Kitchen Knife	721
18	212	Bottle	681
19	312	Pipe/Metal Pipe	659
20	101	Revolver	644
21	304	Club/Bat	635
22	302	Blunt Instrument	530
23	215	Machete	524
24	311	Hammer	511
25	506	Fire	497

Queries (FactVictimAnalysis)

```
-- 4. Average victim age per crime type
SELECT
    ct.CrimeCode      AS [Crime Code],
    ct.CrimeCodeDescription AS [Crime Type],
    SUM(f.SumVictimAge) * 1.0 / NULLIF(SUM(f.VictimCount),0) AS [Avg Victim Age]
FROM FactVictimAnalysis f
JOIN DimCrimeType ct ON f.CrimeTypeKey = ct.CrimeKey
GROUP BY
    ct.CrimeCode,
    ct.CrimeCodeDescription
ORDER BY
    [Avg Victim Age] DESC;
```

Results		Messages	
	Crime Code	Crime Type	Avg Victim Age
1	453	Drunk Roll - Attempt	58.000000000000
2	432	Blocking Door Induction Center	55.000000000000
3	926	Train Wrecking	50.000000000000
4	451	Purse Snatching - Attempt	48.400000000000
5	347	Grand Theft / Insurance Fraud	46.125000000000
6	351	Purse Snatching	45.442857142857
7	933	Prowler	44.312101910828
8	450	Theft From Person - Attempt	44.022727272727
9	901	Violation Of Restraining Order	43.062185856224
10	902	Violation Of Temporary Restr...	42.715151515151
11	900	Violation Of Court Order	42.224018973695
12	624	Battery - Simple Assault	41.944273448890
13	903	Contempt Of Court	41.888943010228
14	421	Theft From Motor Vehicle - At...	41.733750000000
15	354	Theft Of Identity	41.299314044499
16	331	Theft From Motor Vehicle - Gr...	41.041348019632
17	886	Disturbing The Peace	40.940444444444
18	625	Other Assault	40.136227544910
19	520	Vehicle - Attempt Stolen	39.968514531754
20	485	Bike - Attempted Stolen	39.800000000000
21	924	Telephone Property - Damage	38.750000000000
22	930	Criminal Threats - No Weapo...	38.591544470330
23	251	Shots Fired At Inhabited Dwel...	38.533632286995
24	626	Intimate Partner - Simple Assa...	38.022637238256
25	661	Unauthorized Computer Access	37.700000000000

Queries (FactCrimeTime)

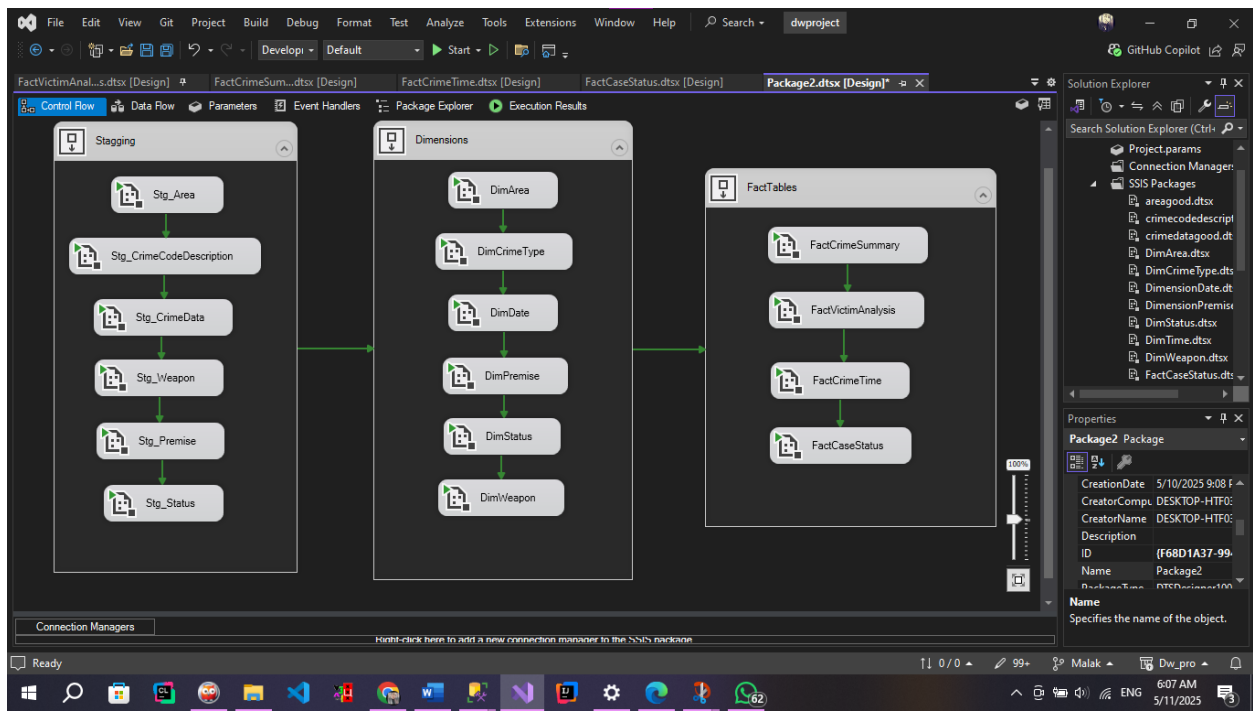
```
-- 5. Crime distribution by time of day
SELECT
    t.PeriodOfDay      AS [Time of Day],
    SUM(f.CrimeCount) AS TotalCrimes
FROM FactCrimeTime f
JOIN DimTime t ON f.TimeKey = t.TimeKey
GROUP BY
    t.PeriodOfDay
ORDER BY
    t.PeriodOfDay
```

Results			Messages		
	Time of Day	TotalCrimes			
1	Afternoon	191688			
2	Evening	140131			
3	Morning	55436			
4	Night	55188			

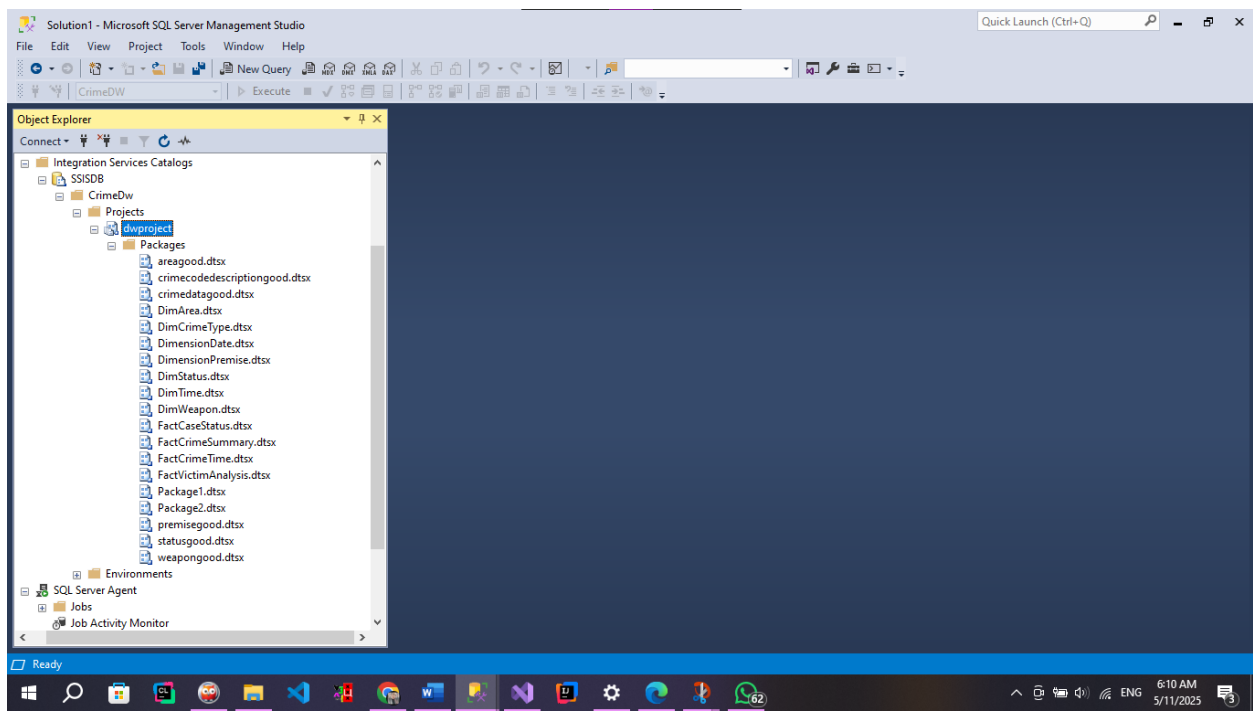
Queries (FactCaseStatus)

```
-- 6. Ratio of status cases
WITH StatusTotals AS (
    SELECT
        s.StatusDescription AS [Status],
        SUM(f.CaseCount) AS CountPerStatus
    FROM FactCaseStatus f
    JOIN DimStatus s ON f.StatusKey = s.StatusKey
    GROUP BY
        s.StatusDescription
)
SELECT
    st.Status,
    st.CountPerStatus,
    st.CountPerStatus * 1.0 / SUM(st.CountPerStatus) OVER() AS [Ratio]
FROM StatusTotals st;
```

Results		Messages	
	Status	CountPerStatus	Ratio
1	Adult Arrest	35721	0.062493220760
2	Adult Other	31119	0.054442107915
3	Invest Cont	502956	0.879912106060
4	Juv Arrest	1258	0.002200847448
5	Juv Other	541	0.000946469371
6	Unknown	3	0.000005248443



The Deployed Packages



The Schedule

File Edit View Project Tools Window Help

Connect New Query Execute

Object Explorer

DESKTOP-HTF03JC (SQL Server 16.0.1135.2 - DESKTOP-HTF03JC)

- Databases
- Security
- Server Objects
- Replication
- Always On High Availability
- Management
- Integration Services Catalogs
- SSISDB
 - CrimeDw
 - Projects
 - dwproject
 - Packages
 - areagood.dtsx
 - crimcodedescriptiongood.dtsx
 - crimedatagood.dtsx
 - DimArea.dtsx
 - DimCrimeType.dtsx
 - DimensionDate.dtsx
 - DimensionPremise.dtsx
 - DimStatus.dtsx
 - DimTime.dtsx
 - DimWeapon.dtsx
 - FactCaseStatus.dtsx
 - FactCrimeSummary.dtsx
 - FactCrimeTime.dtsx
 - FactVictimAnalysis.dtsx
 - Package1.dtsx
 - Package2.dtsx

Select a page

- General
- Steps
- Schedules
- Alerts
- Notifications
- Targets

Script Help

Schedule list:

ID	Name	Enabled	Description
New	CrimeDw	Yes	Occurs every 3 day(s) at 12:00:00 AM, S

Connection

Server: DESKTOP-HTF03JC

Connection: DESKTOP-HTF03JC/user

[View connection properties](#)

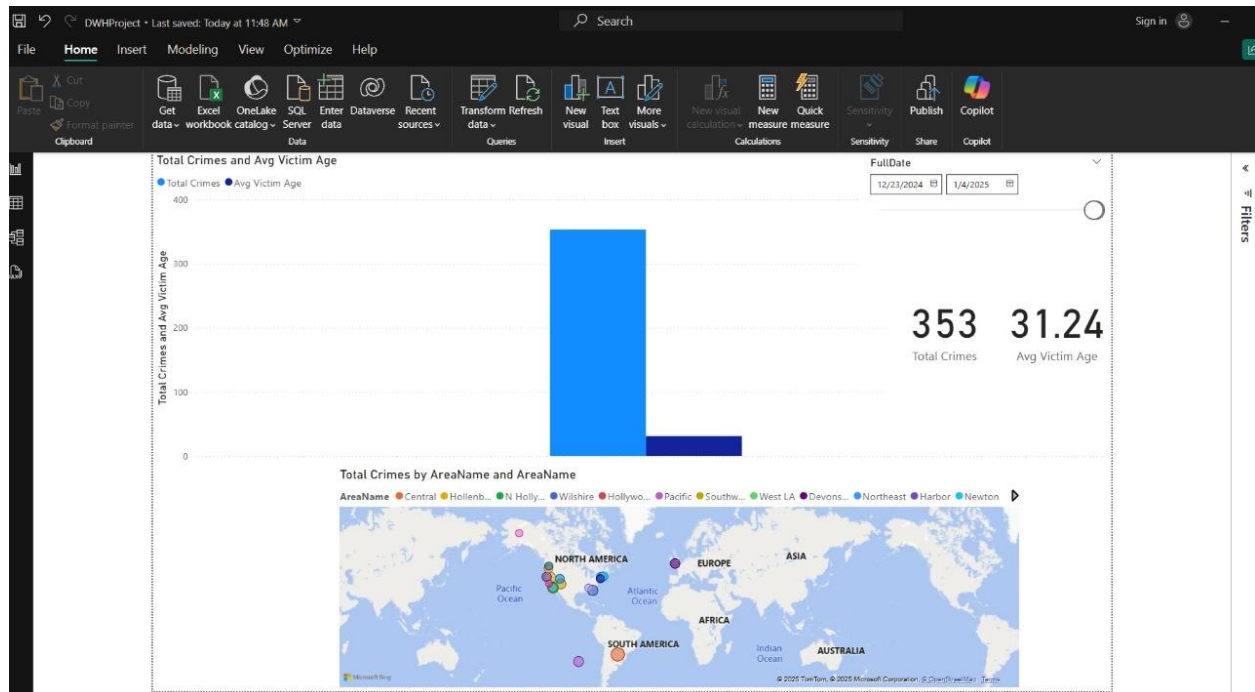
Progress

Ready

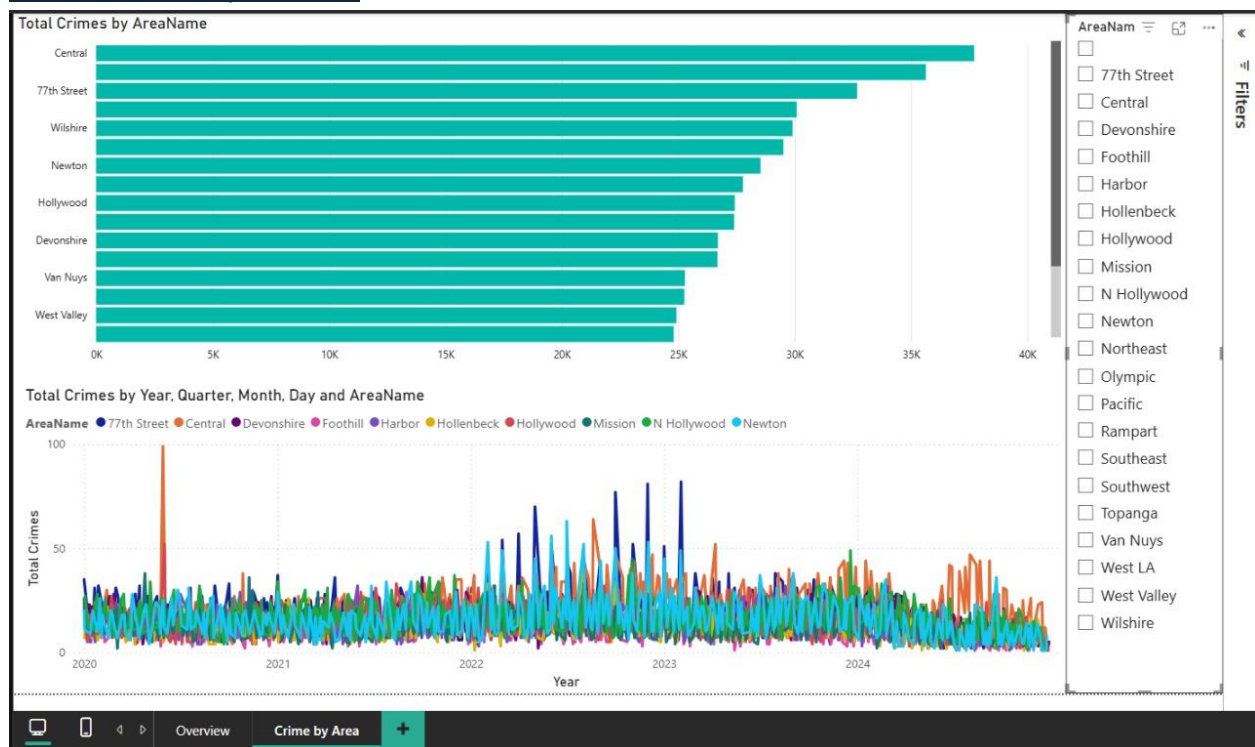
New... Pick... Edit Remove

OK Cancel

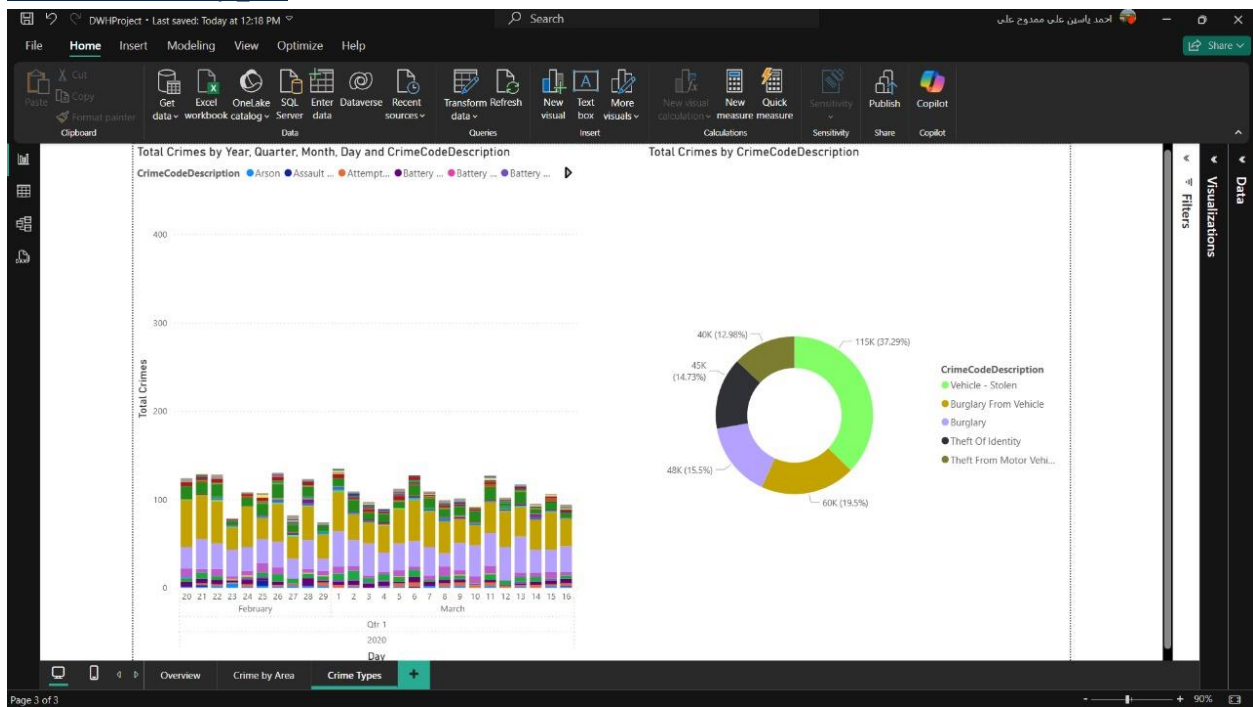
The Bouns 1- Overview



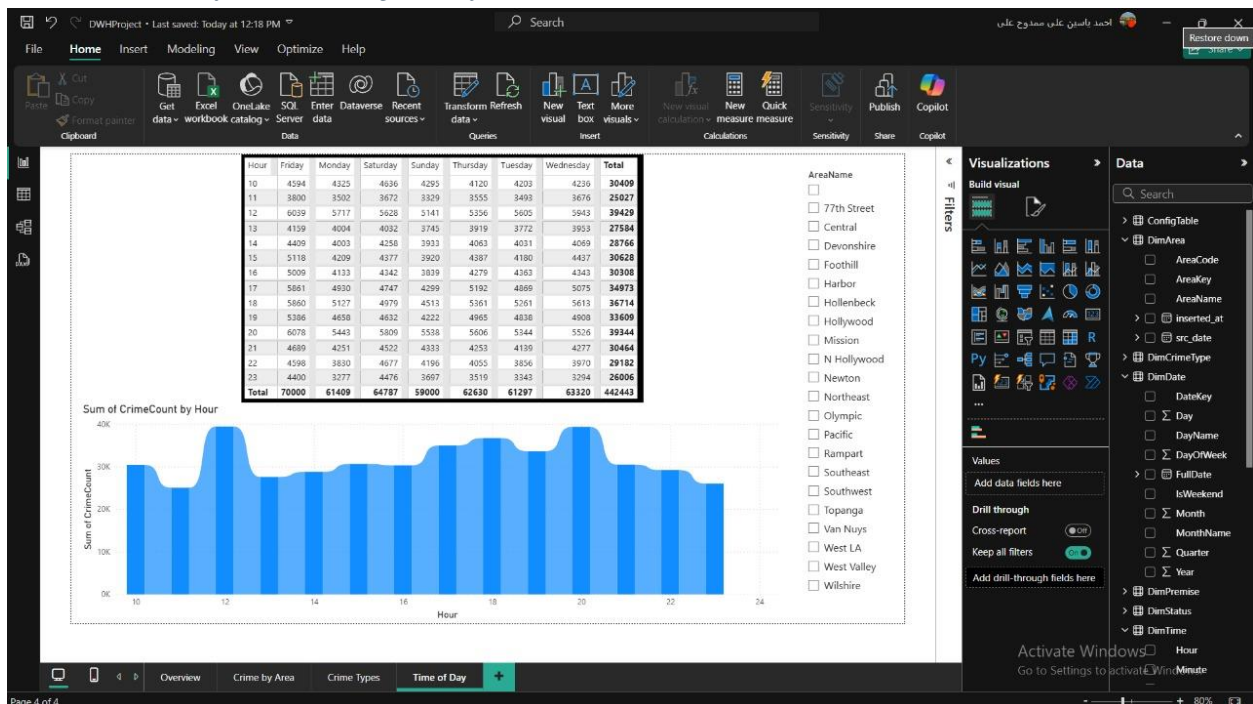
2- Crime by Area



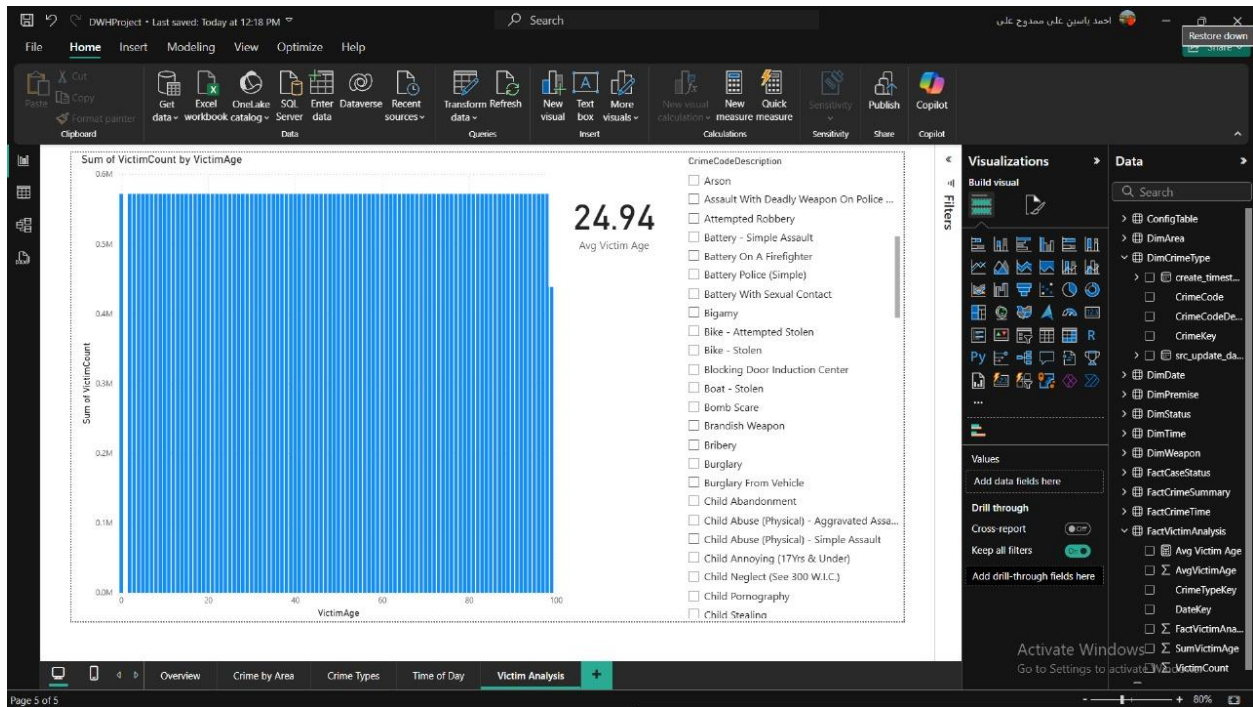
3- Crime Type



4- Crime by Time of Day



5- Victim Analysis



6- Case Status

